

1. The application must set the transfer size and packet count fields in the endpoint-specific registers and enable the endpoint to transmit the data.
2. The application must also write the required data to the associated transmit FIFO for the endpoint.
3. Every time the application writes a packet to the transmit FIFO, the transfer size for that endpoint is decremented by the packet size. The data are fetched from application memory until the transfer size for the endpoint becomes 0.
4. When an IN token is received for a periodic endpoint, the core transmits the data in the FIFO, if available. If the complete data payload (complete packet, in dedicated FIFO mode) for the frame is not present in the FIFO, then the core generates an IN token received when TxFIFO empty interrupt for the endpoint.
  - A zero-length data packet is transmitted on the USB for isochronous IN endpoints
  - A NAK handshake is transmitted on the USB for interrupt IN endpoints
5. The packet count for the endpoint is decremented by 1 under the following conditions:
  - For isochronous endpoints, when a zero- or non-zero-length data packet is transmitted
  - For interrupt endpoints, when an ACK handshake is transmitted
  - When the transfer size and packet count are both 0, the transfer completed interrupt for the endpoint is generated and the endpoint enable is cleared.
6. At the “Periodic frame Interval” (controlled by PFIVL in OTG\_FS\_DCFG), when the core finds non-empty any of the isochronous IN endpoint FIFOs scheduled for the current frame non-empty, the core generates an IISOIXFR interrupt in OTG\_FS\_GINTSTS.

#### Application programming sequence

1. Program the OTG\_FS\_DIEPCTLx register with the endpoint characteristics and set the CNAK and EPENA bits.
2. Write the data to be transmitted in the next frame to the transmit FIFO.
3. Asserting the ITTXFE interrupt (in OTG\_FS\_DIEPINTx) indicates that the application has not yet written all data to be transmitted to the transmit FIFO.
4. If the interrupt endpoint is already enabled when this interrupt is detected, ignore the interrupt. If it is not enabled, enable the endpoint so that the data can be transmitted on the next IN token attempt.
5. Asserting the XFRC interrupt (in OTG\_FS\_DIEPINTx) with no ITTXFE interrupt in OTG\_FS\_DIEPINTx indicates the successful completion of an isochronous IN transfer. A read to the OTG\_FS\_DIEPTSIZx register must give transfer size = 0 and packet count = 0, indicating all data were transmitted on the USB.
6. Asserting the XFRC interrupt (in OTG\_FS\_DIEPINTx), with or without the ITTXFE interrupt (in OTG\_FS\_DIEPINTx), indicates the successful completion of an interrupt IN transfer. A read to the OTG\_FS\_DIEPTSIZx register must give transfer size = 0 and packet count = 0, indicating all data were transmitted on the USB.
7. Asserting the incomplete isochronous IN transfer (IISOIXFR) interrupt in OTG\_FS\_GINTSTS with none of the aforementioned interrupts indicates the core did not receive at least 1 periodic IN token in the current frame.

#### • Incomplete isochronous IN data transfers

This section describes what the application must do on an incomplete isochronous IN data transfer.

**Internal data flow**

1. An isochronous IN transfer is treated as incomplete in one of the following conditions:
  - a) The core receives a corrupted isochronous IN token on at least one isochronous IN endpoint. In this case, the application detects an incomplete isochronous IN transfer interrupt (IISOIXFR in OTG\_FS\_GINTSTS).
  - b) The application is slow to write the complete data payload to the transmit FIFO and an IN token is received before the complete data payload is written to the FIFO. In this case, the application detects an IN token received when TxFIFO empty interrupt in OTG\_FS\_DIEPINTx. The application can ignore this interrupt, as it eventually results in an incomplete isochronous IN transfer interrupt (IISOIXFR in OTG\_FS\_GINTSTS) at the end of periodic frame.  
The core transmits a zero-length data packet on the USB in response to the received IN token.
2. The application must stop writing the data payload to the transmit FIFO as soon as possible.
3. The application must set the NAK bit and the disable bit for the endpoint.
4. The core disables the endpoint, clears the disable bit, and asserts the Endpoint Disable interrupt for the endpoint.

**Application programming sequence**

1. The application can ignore the IN token received when TxFIFO empty interrupt in OTG\_FS\_DIEPINTx on any isochronous IN endpoint, as it eventually results in an incomplete isochronous IN transfer interrupt (in OTG\_FS\_GINTSTS).
2. Assertion of the incomplete isochronous IN transfer interrupt (in OTG\_FS\_GINTSTS) indicates an incomplete isochronous IN transfer on at least one of the isochronous IN endpoints.
3. The application must read the Endpoint Control register for all isochronous IN endpoints to detect endpoints with incomplete IN data transfers.
4. The application must stop writing data to the Periodic Transmit FIFOs associated with these endpoints on the AHB.
5. Program the following fields in the OTG\_FS\_DIEPCTLx register to disable the endpoint:
  - SNAK = 1 in OTG\_FS\_DIEPCTLx
  - EPDIS = 1 in OTG\_FS\_DIEPCTLx
6. The assertion of the Endpoint Disabled interrupt in OTG\_FS\_DIEPINTx indicates that the core has disabled the endpoint.
  - At this point, the application must flush the data in the associated transmit FIFO or overwrite the existing data in the FIFO by enabling the endpoint for a new transfer in the next microframe. To flush the data, the application must use the OTG\_FS\_GRSTCTL register.

- **Stalling non-isochronous IN endpoints**

This section describes how the application can stall a non-isochronous endpoint.

**Application programming sequence**

1. Disable the IN endpoint to be stalled. Set the STALL bit as well.
2. EPDIS = 1 in OTG\_FS\_DIEPCTLx, when the endpoint is already enabled
  - STALL = 1 in OTG\_FS\_DIEPCTLx
  - The STALL bit always takes precedence over the NAK bit
3. Assertion of the Endpoint Disabled interrupt (in OTG\_FS\_DIEPINTx) indicates to the application that the core has disabled the specified endpoint.
4. The application must flush the non-periodic or periodic transmit FIFO, depending on the endpoint type. In case of a non-periodic endpoint, the application must re-enable the other non-periodic endpoints that do not need to be stalled, to transmit data.
5. Whenever the application is ready to end the STALL handshake for the endpoint, the STALL bit must be cleared in OTG\_FS\_DIEPCTLx.
6. If the application sets or clears a STALL bit for an endpoint due to a SetFeature.Endpoint Halt command or ClearFeature.Endpoint Halt command, the STALL bit must be set or cleared before the application sets up the Status stage transfer on the control endpoint.

#### Special case: stalling the control OUT endpoint

The core must stall IN/OUT tokens if, during the data stage of a control transfer, the host sends more IN/OUT tokens than are specified in the SETUP packet. In this case, the application must enable the ITTXFE interrupt in OTG\_FS\_DIEPINTx and the OTEPDIS interrupt in OTG\_FS\_DOEPINTx during the data stage of the control transfer, after the core has transferred the amount of data specified in the SETUP packet. Then, when the application receives this interrupt, it must set the STALL bit in the corresponding endpoint control register, and clear this interrupt.

### 22.17.7 Worst case response time

When the OTG\_FS controller acts as a device, there is a worst case response time for any tokens that follow an isochronous OUT. This worst case response time depends on the AHB clock frequency.

The core registers are in the AHB domain, and the core does not accept another token before updating these register values. The worst case is for any token following an isochronous OUT, because for an isochronous transaction, there is no handshake and the next token could come sooner. This worst case value is 7 PHY clocks when the AHB clock is the same as the PHY clock. When the AHB clock is faster, this value is smaller.

If this worst case condition occurs, the core responds to bulk/interrupt tokens with a NAK and drops isochronous and SETUP tokens. The host interprets this as a timeout condition for SETUP and retries the SETUP packet. For isochronous transfers, the Incomplete isochronous IN transfer interrupt (IISOIXFR) and Incomplete isochronous OUT transfer interrupt (IISOOXFR) inform the application that isochronous IN/OUT packets were dropped.

#### Choosing the value of TRDT in OTG\_FS\_GUSBCFG

The value in TRDT (OTG\_FS\_GUSBCFG) is the time it takes for the MAC, in terms of PHY clocks after it has received an IN token, to get the FIFO status, and thus the first data from the PFC block. This time involves the synchronization delay between the PHY and AHB clocks. The worst case delay for this is when the AHB clock is the same as the PHY clock. In this case, the delay is 5 clocks.

Once the MAC receives an IN token, this information (token received) is synchronized to the AHB clock by the PFC (the PFC runs on the AHB clock). The PFC then reads the data from the SPRAM and writes them into the dual clock source buffer. The MAC then reads the data out of the source buffer (4 deep).

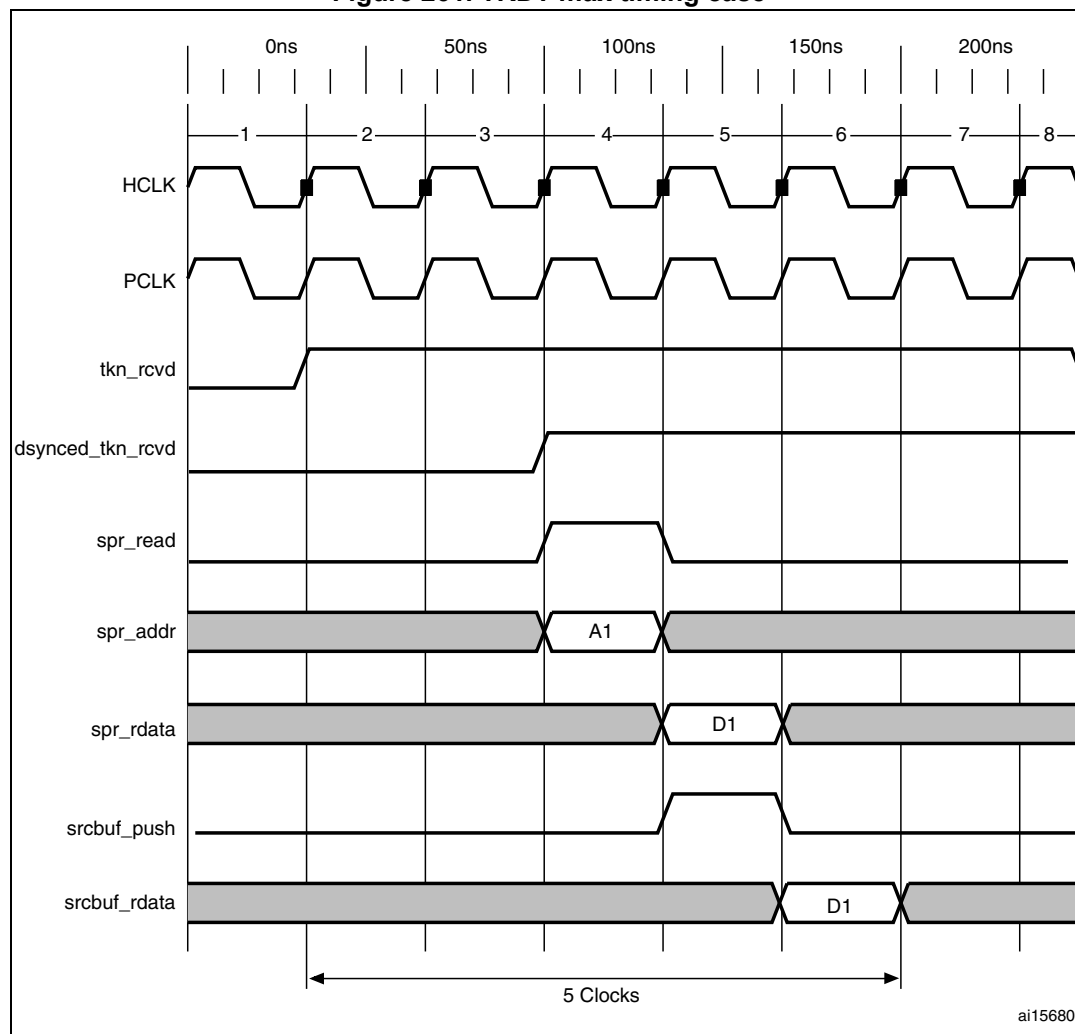
If the AHB is running at a higher frequency than the PHY, the application can use a smaller value for TRDT (in OTG\_FS\_GUSBCFG).

[Figure 261](#) has the following signals:

- `tkn_rcvd`: Token received information from MAC to PFC
- `dynced_tkn_rcvd`: Doubled sync `tkn_rcvd`, from PCLK to HCLK domain
- `spr_read`: Read to SPRAM
- `spr_addr`: Address to SPRAM
- `spr_rdata`: Read data from SPRAM
- `srcbuf_push`: Push to the source buffer
- `srcbuf_rdata`: Read data from the source buffer. Data seen by MAC

Refer to [Table 132: TRDT values](#) for the values of TRDT versus AHB clock frequency.

**Figure 261. TRDT max timing case**



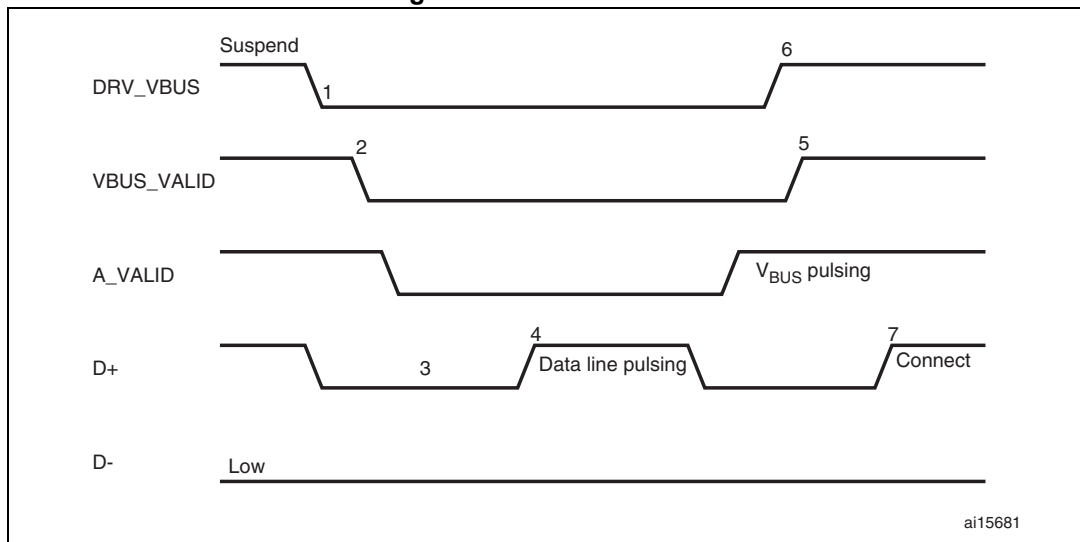
### 22.17.8 OTG programming model

The OTG\_FS controller is an OTG device supporting HNP and SRP. When the core is connected to an “A” plug, it is referred to as an A-device. When the core is connected to a “B” plug it is referred to as a B-device. In host mode, the OTG\_FS controller turns off  $V_{BUS}$  to conserve power. SRP is a method by which the B-device signals the A-device to turn on  $V_{BUS}$  power. A device must perform both data-line pulsing and  $V_{BUS}$  pulsing, but a host can detect either data-line pulsing or  $V_{BUS}$  pulsing for SRP. HNP is a method by which the B-device negotiates and switches to host role. In Negotiated mode after HNP, the B-device suspends the bus and reverts to the device role.

#### A-device session request protocol

The application must set the SRP-capable bit in the Core USB configuration register. This enables the OTG\_FS controller to detect SRP as an A-device.

Figure 262. A-device SRP



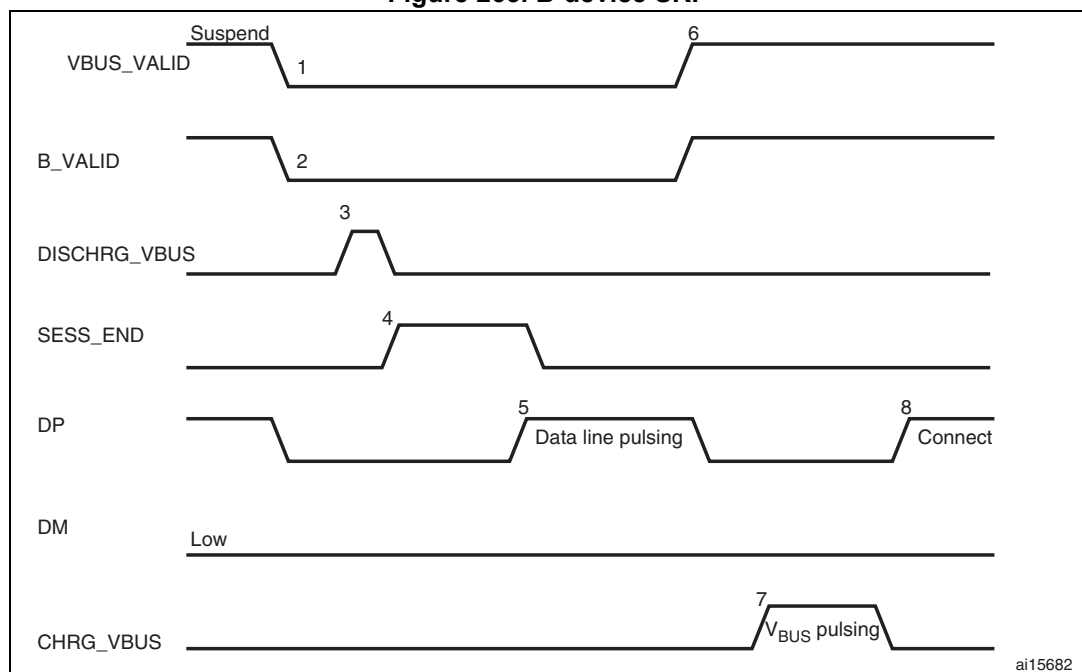
1. DRV\_VBUS =  $V_{BUS}$  drive signal to the PHY  
 VBUS\_VALID =  $V_{BUS}$  valid signal from PHY  
 A\_VALID = A-peripheral  $V_{BUS}$  level signal to PHY  
 D+ = Data plus line  
 D- = Data minus line

1. To save power, the application suspends and turns off port power when the bus is idle by writing the port suspend and port power bits in the host port control and status register.
  2. PHY indicates port power off by deasserting the VBUS\_VALID signal.
  3. The device must detect SE0 for at least 2 ms to start SRP when V<sub>BUS</sub> power is off.
  4. To initiate SRP, the device turns on its data line pull-up resistor for 5 to 10 ms. The OTG\_FS controller detects data-line pulsing.
  5. The device drives V<sub>BUS</sub> above the A-device session valid (2.0 V minimum) for V<sub>BUS</sub> pulsing.
- The OTG\_FS controller interrupts the application on detecting SRP. The Session request detected bit is set in Global interrupt status register (SRQINT set in OTG\_FS\_GINTSTS).
6. The application must service the Session request detected interrupt and turn on the port power bit by writing the port power bit in the host port control and status register. The PHY indicates port power-on by asserting the VBUS\_VALID signal.
  7. When the USB is powered, the device connects, completing the SRP process.

### B-device session request protocol

The application must set the SRP-capable bit in the Core USB configuration register. This enables the OTG\_FS controller to initiate SRP as a B-device. SRP is a means by which the OTG\_FS controller can request a new session from the host.

**Figure 263. B-device SRP**



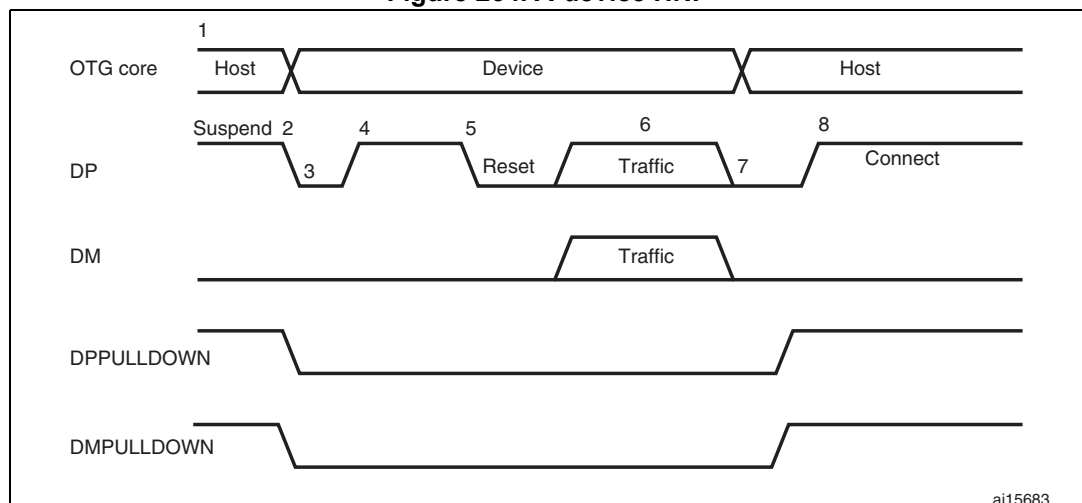
1. VBUS\_VALID = V<sub>BUS</sub> valid signal from PHY  
 B\_VALID = B-peripheral valid session to PHY  
 DISCHRG\_VBUS = discharge signal to PHY  
 SESS\_END = session end signal to PHY  
 CHRG\_VBUS = charge V<sub>BUS</sub> signal to PHY  
 DP = Data plus line  
 DM = Data minus line

1. To save power, the host suspends and turns off port power when the bus is idle.  
The OTG\_FS controller sets the early suspend bit in the Core interrupt register after 3 ms of bus idleness. Following this, the OTG\_FS controller sets the USB suspend bit in the Core interrupt register.  
The OTG\_FS controller informs the PHY to discharge  $V_{BUS}$ .
2. The PHY indicates the session's end to the device. This is the initial condition for SRP.  
The OTG\_FS controller requires 2 ms of SE0 before initiating SRP.  
For a USB 1.1 full-speed serial transceiver, the application must wait until  $V_{BUS}$  discharges to 0.2 V after BSVLD (in OTG\_FS\_GOTGCTL) is deasserted. This discharge time can be obtained from the transceiver vendor and varies from one transceiver to another.
3. The USB OTG core informs the PHY to speed up  $V_{BUS}$  discharge.
4. The application initiates SRP by writing the session request bit in the OTG Control and status register. The OTG\_FS controller perform data-line pulsing followed by  $V_{BUS}$  pulsing.
5. The host detects SRP from either the data-line or  $V_{BUS}$  pulsing, and turns on  $V_{BUS}$ .  
The PHY indicates  $V_{BUS}$  power-on to the device.
6. The OTG\_FS controller performs  $V_{BUS}$  pulsing.  
The host starts a new session by turning on  $V_{BUS}$ , indicating SRP success. The OTG\_FS controller interrupts the application by setting the session request success status change bit in the OTG interrupt status register. The application reads the session request success bit in the OTG control and status register.
7. When the USB is powered, the OTG\_FS controller connects, completing the SRP process.

### A-device host negotiation protocol

HNP switches the USB host role from the A-device to the B-device. The application must set the HNP-capable bit in the Core USB configuration register to enable the OTG\_FS controller to perform HNP as an A-device.

**Figure 264. A-device HNP**



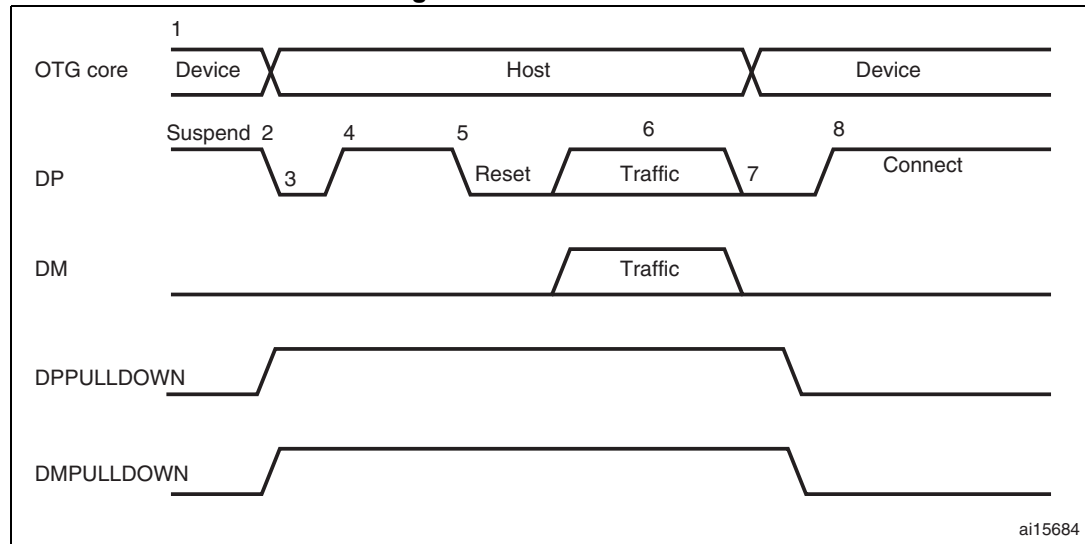
1. DPPULLDOWN = signal from core to PHY to enable/disable the pull-down on the DP line inside the PHY.  
DMPULLDOWN = signal from core to PHY to enable/disable the pull-down on the DM line inside the PHY.

1. The OTG\_FS controller sends the B-device a SetFeature b\_hnp\_enable descriptor to enable HNP support. The B-device's ACK response indicates that the B-device supports HNP. The application must set host Set HNP Enable bit in the OTG Control and status register to indicate to the OTG\_FS controller that the B-device supports HNP.
2. When it has finished using the bus, the application suspends by writing the Port suspend bit in the host port control and status register.
3. When the B-device observes a USB suspend, it disconnects, indicating the initial condition for HNP. The B-device initiates HNP only when it must switch to the host role; otherwise, the bus continues to be suspended.  
The OTG\_FS controller sets the host negotiation detected interrupt in the OTG interrupt status register, indicating the start of HNP.  
The OTG\_FS controller deasserts the DM pull down and DM pull down in the PHY to indicate a device role. The PHY enables the OTG\_FS\_DP pull-up resistor to indicate a connect for B-device.  
The application must read the current mode bit in the OTG Control and status register to determine device mode operation.
4. The B-device detects the connection, issues a USB reset, and enumerates the OTG\_FS controller for data traffic.
5. The B-device continues the host role, initiating traffic, and suspends the bus when done.  
The OTG\_FS controller sets the early suspend bit in the Core interrupt register after 3 ms of bus idleness. Following this, the OTG\_FS controller sets the USB Suspend bit in the Core interrupt register.
6. In Negotiated mode, the OTG\_FS controller detects the suspend, disconnects, and switches back to the host role. The OTG\_FS controller asserts the DM pull down and DM pull down in the PHY to indicate its assumption of the host role.
7. The OTG\_FS controller sets the Connector ID status change interrupt in the OTG Interrupt Status register. The application must read the connector ID status in the OTG Control and Status register to determine the OTG\_FS controller operation as an A-device. This indicates the completion of HNP to the application. The application must read the Current mode bit in the OTG control and status register to determine host mode operation.
8. The B-device connects, completing the HNP process.

## B-device host negotiation protocol

HNP switches the USB host role from B-device to A-device. The application must set the HNP-capable bit in the Core USB configuration register to enable the OTG\_FS controller to perform HNP as a B-device.

**Figure 265. B-device HNP**



1. DPPULLDOWN = signal from core to PHY to enable/disable the pull-down on the DP line inside the PHY.  
DMPULLDOWN = signal from core to PHY to enable/disable the pull-down on the DM line inside the PHY.
1. The A-device sends the SetFeature b\_hnp\_enable descriptor to enable HNP support. The OTG\_FS controller's ACK response indicates that it supports HNP. The application must set the device HNP enable bit in the OTG Control and status register to indicate HNP support.  
The application sets the HNP request bit in the OTG Control and status register to indicate to the OTG\_FS controller to initiate HNP.
2. When it has finished using the bus, the A-device suspends by writing the Port suspend bit in the host port control and status register.  
The OTG\_FS controller sets the Early suspend bit in the Core interrupt register after 3 ms of bus idleness. Following this, the OTG\_FS controller sets the USB suspend bit in the Core interrupt register.  
The OTG\_FS controller disconnects and the A-device detects SE0 on the bus, indicating HNP. The OTG\_FS controller asserts the DP pull down and DM pull down in the PHY to indicate its assumption of the host role.  
The A-device responds by activating its OTG\_FS\_DP pull-up resistor within 3 ms of detecting SE0. The OTG\_FS controller detects this as a connect.  
The OTG\_FS controller sets the host negotiation success status change interrupt in the OTG Interrupt status register, indicating the HNP status. The application must read the host negotiation success bit in the OTG Control and status register to determine host negotiation success. The application must read the current Mode bit in the Core interrupt register (OTG\_FS\_GINTSTS) to determine host mode operation.
3. The application sets the reset bit (PRST in OTG\_FS\_HPRT) and the OTG\_FS controller issues a USB reset and enumerates the A-device for data traffic.

4. The OTG\_FS controller continues the host role of initiating traffic, and when done, suspends the bus by writing the Port suspend bit in the host port control and status register.
5. In Negotiated mode, when the A-device detects a suspend, it disconnects and switches back to the host role. The OTG\_FS controller deasserts the DP pull down and DM pull down in the PHY to indicate the assumption of the device role.
6. The application must read the current mode bit in the Core interrupt (OTG\_FS\_GINTSTS) register to determine the host mode operation.
7. The OTG\_FS controller connects, completing the HNP process.

## 23 Debug support (DBG)

### 23.1 Overview

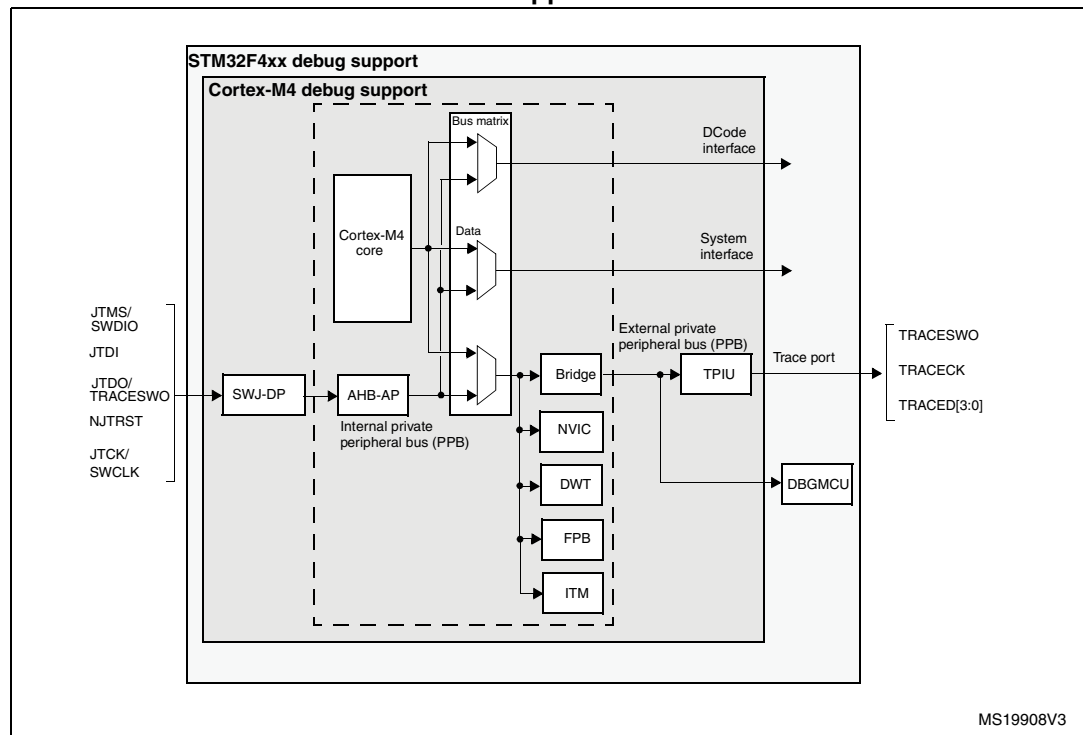
The STM32F411xC/E is built around a Cortex<sup>®</sup>-M4 with FPU core which contains hardware extensions for advanced debugging features. The debug extensions allow the core to be stopped either on a given instruction fetch (breakpoint) or data access (watchpoint). When stopped, the core's internal state and the system's external state may be examined. Once examination is complete, the core and the system may be restored and program execution resumed.

The debug features are used by the debugger host when connecting to and debugging the STM32F411xC/E MCUs.

Two interfaces for debug are available:

- Serial wire
- JTAG debug port

**Figure 266. Block diagram of STM32 MCU and Cortex<sup>®</sup>-M4 with FPU-level debug support**



**Note:** The debug features embedded in the Cortex<sup>®</sup>-M4 with FPU core are a subset of the Arm<sup>®</sup> CoreSight Design Kit.

The Arm® Cortex®-M4 with FPU core provides integrated on-chip debug support. It is comprised of:

- SWJ-DP: Serial wire / JTAG debug port
- AHP-AP: AHB access port
- ITM: Instrumentation trace macrocell
- FPB: Flash patch breakpoint
- DWT: Data watchpoint trigger
- TPUI: Trace port unit interface (available on larger packages, where the corresponding pins are mapped)
- ETM: Embedded Trace Macrocell (available on larger packages, where the corresponding pins are mapped)

It also includes debug features dedicated to the STM32F411xC/E:

- Flexible debug pinout assignment
- MCU debug box (support for low-power modes, control over peripheral clocks, etc.)

*Note:* For further information on debug functionality supported by the Arm® Cortex®-M4 with FPU core, refer to the Cortex®-M4 with FPU -r0p1 Technical Reference Manual and to the CoreSight Design Kit-r0p1 TRM (see [Section 23.2: Reference Arm® documentation](#)).

## 23.2 Reference Arm® documentation

- Cortex®-M4 with FPU r0p1 Technical Reference Manual (TRM)  
(see Related documents on page 1)
- Arm® Debug Interface V5
- Arm® CoreSight Design Kit revision r0p1 Technical Reference Manual

## 23.3 SWJ debug port (serial wire and JTAG)

The core of the STM32F411xC/E integrates the Serial Wire / JTAG Debug Port (SWJ-DP). It is an Arm® standard CoreSight debug port that combines a JTAG-DP (5-pin) interface and a SW-DP (2-pin) interface.

- The JTAG Debug Port (JTAG-DP) provides a 5-pin standard JTAG interface to the AHP-AP port.
- The Serial Wire Debug Port (SW-DP) provides a 2-pin (clock + data) interface to the AHP-AP port.

In the SWJ-DP, the two JTAG pins of the SW-DP are multiplexed with some of the five JTAG pins of the JTAG-DP.

Figure 267. SWJ debug port

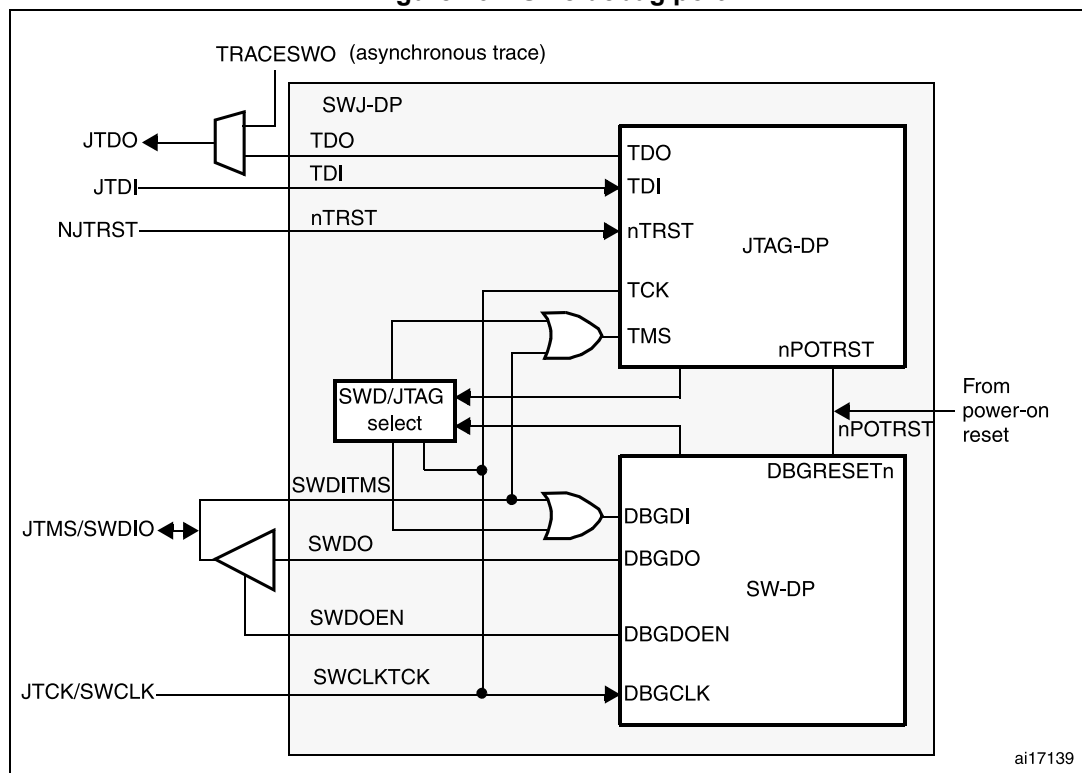


Figure 267 shows that the asynchronous TRACE output (TRACESWO) is multiplexed with TDO. This means that the asynchronous trace can only be used with SW-DP, not JTAG-DP.

### 23.3.1 Mechanism to select the JTAG-DP or the SW-DP

By default, the JTAG-Debug Port is active.

If the debugger host wants to switch to the SW-DP, it must provide a dedicated JTAG sequence on TMS/TCK (respectively mapped to SWDIO and SWCLK) which disables the JTAG-DP and enables the SW-DP. This way it is possible to activate the SWDP using only the SWCLK and SWDIO pins.

This sequence is:

1. Send more than 50 TCK cycles with TMS (SWDIO) = 1
2. Send the 16-bit sequence on TMS (SWDIO) = 0111100111100111 (MSB transmitted first)
3. Send more than 50 TCK cycles with TMS (SWDIO) = 1

## 23.4 Pinout and debug port pins

The STM32F411xC/E MCUs are available in various packages with different numbers of available pins. As a result, some functionality (ETM) related to pin availability may differ between packages.

### 23.4.1 SWJ debug port pins

Five pins are used as outputs from the STM32F411xC/E for the SWJ-DP as *alternate functions* of general-purpose I/Os. These pins are available on all packages.

**Table 135. SWJ debug port pins**

| SWJ-DP pin name | JTAG debug port |                          | SW debug port |                                    | Pin assignment |
|-----------------|-----------------|--------------------------|---------------|------------------------------------|----------------|
|                 | Type            | Description              | Type          | Debug assignment                   |                |
| JTMS/SWDIO      | I               | JTAG Test Mode Selection | IO            | Serial Wire Data Input/Output      | PA13           |
| JTCK/SWCLK      | I               | JTAG Test Clock          | I             | Serial Wire Clock                  | PA14           |
| JTDI            | I               | JTAG Test Data Input     | -             | -                                  | PA15           |
| JTDO/TRACESWO   | O               | JTAG Test Data Output    | -             | TRACESWO if async trace is enabled | PB3            |
| NJTRST          | I               | JTAG Test nReset         | -             | -                                  | PB4            |

### 23.4.2 Flexible SWJ-DP pin assignment

After RESET (SYSRESETn or PORESETn), all five pins used for the SWJ-DP are assigned as dedicated pins immediately usable by the debugger host (note that the trace outputs are not assigned except if explicitly programmed by the debugger host).

However, the STM32F411xC/E MCU offers the possibility of disabling some or all of the SWJ-DP ports and so, of releasing the associated pins for general-purpose IO (GPIO) usage. For more details on how to disable SWJ-DP port pins, please refer to [Section 8.3.2: I/O pin multiplexer and mapping](#).

**Table 136. Flexible SWJ-DP pin assignment**

| Available debug ports                         | SWJ IO pin assigned |                     |             |            |              |
|-----------------------------------------------|---------------------|---------------------|-------------|------------|--------------|
|                                               | PA13 / JTMS / SWDIO | PA14 / JTCK / SWCLK | PA15 / JTDI | PB3 / JTDO | PB4 / NJTRST |
| Full SWJ (JTAG-DP + SW-DP) - Reset State      | X                   | X                   | X           | X          | X            |
| Full SWJ (JTAG-DP + SW-DP) but without NJTRST | X                   | X                   | X           | X          |              |
| JTAG-DP Disabled and SW-DP Enabled            | X                   | X                   |             |            |              |
| JTAG-DP Disabled and SW-DP Disabled           | Released            |                     |             |            |              |

**Note:** When the APB bridge write buffer is full, it takes one extra APB cycle when writing the GPIO\_AFR register. This is because the deactivation of the JTAGSW pins is done in two cycles to guarantee a clean level on the nTRST and TCK input signals of the core.

- Cycle 1: the JTAGSW input signals to the core are tied to 1 or 0 (to 1 for nTRST, TDI and TMS, to 0 for TCK)
- Cycle 2: the GPIO controller takes the control signals of the SWJTAG IO pins (like controls of direction, pull-up/down, Schmitt trigger activation, etc.).

### 23.4.3 Internal pull-up and pull-down on JTAG pins

It is necessary to ensure that the JTAG input pins are not floating since they are directly connected to flip-flops to control the debug mode features. Special care must be taken with the SWCLK/TCK pin which is directly connected to the clock of some of these flip-flops.

To avoid any uncontrolled IO levels, the device embeds internal pull-ups and pull-downs on the JTAG input pins:

NJTRST: Internal pull-up  
JTDI: Internal pull-up  
JTMS/SWDIO: Internal pull-up  
TCK/SWCLK: Internal pull-down  
Once a JTAG IO is released by the user software, the GPIO controller takes control again. The reset states of the GPIO control registers put the I/Os in the equivalent state:

- NJTRST: AF input pull-up
- JTDI: AF input pull-up
- JTMS/SWDIO: AF input pull-up
- JTCK/SWCLK: AF input pull-down
- JTDO: AF output floating

The software can then use these I/Os as standard GPIOs.

*Note: The JTAG IEEE standard recommends to add pull-ups on TDI, TMS and nTRST but there is no special recommendation for TCK. However, for JTCK, the device needs an integrated pull-down.*

*Having embedded pull-ups and pull-downs removes the need to add external resistors.*

### 23.4.4 Using serial wire and releasing the unused debug pins as GPIOs

To use the serial wire DP to release some GPIOs, the user software must change the GPIO (PA15, PB3 and PB4) configuration mode in the GPIO\_MODER register. This releases PA15, PB3 and PB4 which now become available as GPIOs.

When debugging, the host performs the following actions:

- Under system reset, all SWJ pins are assigned (JTAG-DP + SW-DP).
- Under system reset, the debugger host sends the JTAG sequence to switch from the JTAG-DP to the SW-DP.
- Still under system reset, the debugger sets a breakpoint on vector reset.
- The system reset is released and the Core halts.
- All the debug communications from this point are done using the SW-DP. The other JTAG pins can then be reassigned as GPIOs by the user software.

*Note:* For user software designs, note that:

*To release the debug pins, remember that they will be first configured either in input-pull-up (nTRST, TMS, TDI) or pull-down (TCK) or output tristate (TDO) for a certain duration after reset until the instant when the user software releases the pins.*

*When debug pins (JTAG or SW or TRACE) are mapped, changing the corresponding IO pin configuration in the IOPORT controller has no effect.*

## 23.5 STM32F411xC/E JTAG TAP connection

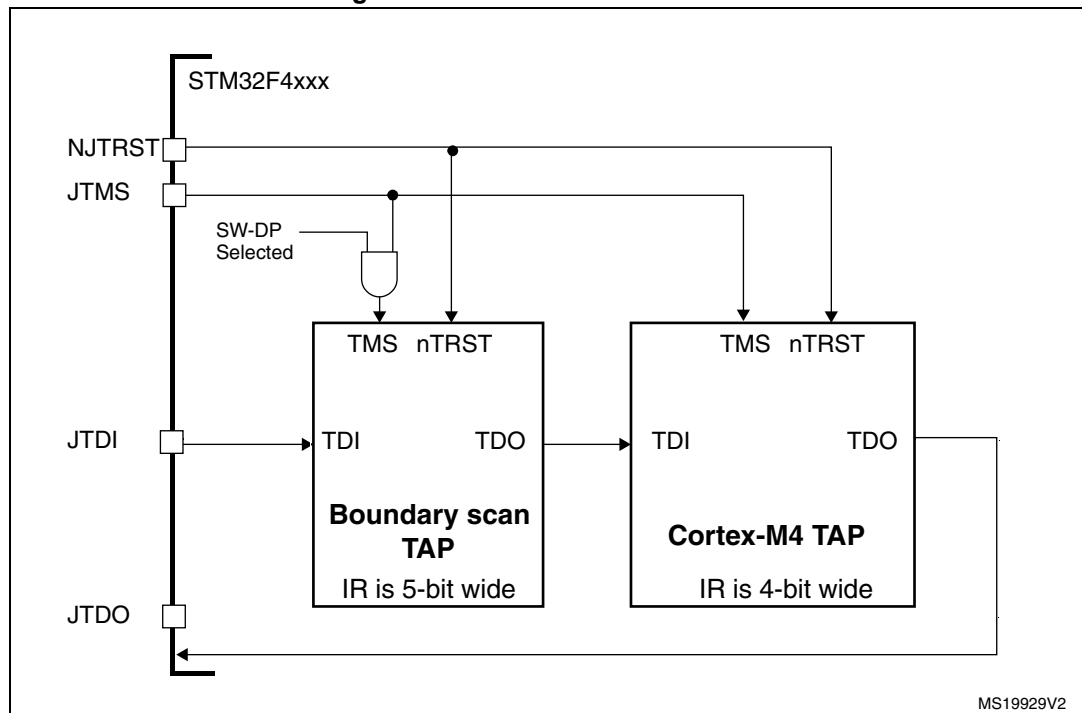
The STM32F411xC/E MCUs integrate two serially connected JTAG TAPs, the boundary scan TAP (IR is 5-bit wide) and the Cortex®-M4 with FPU TAP (IR is 4-bit wide).

To access the TAP of the Cortex®-M4 with FPU for debug purposes:

1. First, it is necessary to shift the BYPASS instruction of the boundary scan TAP.
2. Then, for each IR shift, the scan chain contains 9 bits (=5+4) and the unused TAP instruction must be shifted in using the BYPASS instruction.
3. For each data shift, the unused TAP, which is in BYPASS mode, adds 1 extra data bit in the data scan chain.

*Note:* **Important:** Once Serial-Wire is selected using the dedicated Arm® JTAG sequence, the boundary scan TAP is automatically disabled (JTMS forced high).

Figure 268. JTAG TAP connections



## 23.6 ID codes and locking mechanism

There are several ID codes inside the STM32F411xx MCUs. ST strongly recommends tools designers to lock their debuggers using the MCU DEVICE ID code located in the external PPB memory map at address 0xE0042000.

### 23.6.1 MCU device ID code

The STM32F411xC/E MCUs integrate an MCU ID code. This ID identifies the ST MCU part-number and the die revision. It is part of the DBG\_MCU component and is mapped on the external PPB bus (see [Section 23.16 on page 826](#)). This code is accessible using the JTAG debug port (four to five pins) or the SW debug port (two pins) or by the user software. It is even accessible while the MCU is under system reset.

Only the DEV\_ID[11:0] should be used for identification by the debugger/programmer tools.

#### DBGMCU\_IDCODE

Address: 0xE004 2000

Only 32-bits access supported. Read-only.

|              |    |    |    |              |    |    |    |    |    |    |    |    |    |    |    |
|--------------|----|----|----|--------------|----|----|----|----|----|----|----|----|----|----|----|
| 31           | 30 | 29 | 28 | 27           | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| REV_ID[15:0] |    |    |    |              |    |    |    |    |    |    |    |    |    |    |    |
| r            | r  | r  | r  | r            | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  |
| 15           | 14 | 13 | 12 | 11           | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Reserved     |    |    |    | DEV_ID[11:0] |    |    |    |    |    |    |    |    |    |    |    |
|              |    |    |    | r            | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  |

Bits 31:16 **REV\_ID[15:0]** Revision identifier

This field indicates the revision of the device:

0x1000 = Revision A

Bits 15:12 Reserved, must be kept at reset value.

Bits 11:0 **DEV\_ID[11:0]**: Device identifier

The device ID is 0x431 for the STM32F411xC/E

### 23.6.2 Boundary scan TAP

#### JTAG ID code

The TAP of the STM32F411xC/E BSC (boundary scan) integrates a JTAG ID code equal to 0x06431041 .

### 23.6.3 Cortex®-M4 with FPU TAP

The TAP of the Arm® Cortex®-M4 with FPU integrates a JTAG ID code. This ID code is the Arm® default one and has not been modified. This code is only accessible by the JTAG Debug Port, it is (corresponds to Cortex®-M4 with FPU r0p1, see [Section 23.2](#)).

### 23.6.4 Cortex®-M4 with FPU JEDEC-106 ID code

The Arm® Cortex®-M4 with FPU integrates a JEDEC-106 ID code. It is located in the 4 KB ROM table mapped on the internal PPB bus at address 0xE00FF000\_0xE00FFFFF.

This code is accessible by the JTAG Debug Port (4 to 5 pins) or by the SW Debug Port (two pins) or by the user software.

## 23.7 JTAG debug port

A standard JTAG state machine is implemented with a 4-bit instruction register (IR) and five data registers (for full details, refer to the Cortex®-M4 with FPU r0p1 *Technical Reference Manual (TRM)*, for references, see [Section 23.2](#)).

**Table 137. JTAG debug port data registers**

| IR(3:0) | Data register       | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1111    | BYPASS<br>[1 bit]   | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1110    | IDCODE<br>[32 bits] | ID CODE<br>0x0BA00477 (Arm® Cortex®-M4 with FPU r0p1 ID Code)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 1010    | DPACC<br>[35 bits]  | Debug port access register<br>This initiates a debug port and allows access to a debug port register.<br>– When transferring data IN:<br>Bits 34:3 = DATA[31:0] = 32-bit data to transfer for a write request<br>Bits 2:1 = A[3:2] = 2-bit address of a debug port register.<br>Bit 0 = RnW = Read request (1) or write request (0).<br>– When transferring data OUT:<br>Bits 34:3 = DATA[31:0] = 32-bit data which is read following a read request<br>Bits 2:0 = ACK[2:0] = 3-bit Acknowledge:<br>010 = OK/FAULT<br>001 = WAIT<br>OTHER = reserved<br>Refer to <a href="#">Table 138</a> for a description of the A[3:2] bits |

Table 137. JTAG debug port data registers (continued)

| IR(3:0) | Data register      | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1011    | APACC<br>[35 bits] | <p>Access port access register</p> <p>Initiates an access port and allows access to an access port register.</p> <ul style="list-style-type: none"> <li>When transferring data IN: <ul style="list-style-type: none"> <li>Bits 34:3 = DATA[31:0] = 32-bit data to shift in for a write request</li> <li>Bits 2:1 = A[3:2] = 2-bit address (sub-address AP registers).</li> <li>Bit 0 = RnW= Read request (1) or write request (0).</li> </ul> </li> <li>When transferring data OUT: <ul style="list-style-type: none"> <li>Bits 34:3 = DATA[31:0] = 32-bit data which is read following a read request</li> <li>Bits 2:0 = ACK[2:0] = 3-bit Acknowledge: <ul style="list-style-type: none"> <li>010 = OK/FAULT</li> <li>001 = WAIT</li> <li>OTHER = reserved</li> </ul> </li> </ul> </li> </ul> <p>There are many AP Registers (see AHB-AP) addressed as the combination of:</p> <ul style="list-style-type: none"> <li>The shifted value A[3:2]</li> <li>The current value of the DP SELECT register</li> </ul> |
| 1000    | ABORT<br>[35 bits] | <p>Abort register</p> <ul style="list-style-type: none"> <li>Bits 31:1 = Reserved</li> <li>Bit 0 = DAPABORT: write 1 to generate a DAP abort.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

Table 138. 32-bit debug port registers addressed through the shifted value A[3:2]

| Address | A[3:2] value | Description                                                                                                                                                                                                                                                                                                                                             |
|---------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x0     | 00           | Reserved, must be kept at reset value.                                                                                                                                                                                                                                                                                                                  |
| 0x4     | 01           | <p>DP CTRL/STAT register. Used to:</p> <ul style="list-style-type: none"> <li>Request a system or debug power-up</li> <li>Configure the transfer operation for AP accesses</li> <li>Control the pushed compare and pushed verify operations.</li> <li>Read some status flags (overrun, power-up acknowledges)</li> </ul>                                |
| 0x8     | 10           | <p>DP SELECT register: Used to select the current access port and the active 4-words register window.</p> <ul style="list-style-type: none"> <li>Bits 31:24: APSEL: select the current AP</li> <li>Bits 23:8: reserved</li> <li>Bits 7:4: APBANKSEL: select the active 4-words register window on the current AP</li> <li>Bits 3:0: reserved</li> </ul> |
| 0xC     | 11           | <p>DP RDBUFF register: Used to allow the debugger to get the final result after a sequence of operations (without requesting new JTAG-DP operation)</p>                                                                                                                                                                                                 |

## 23.8 SW debug port

### 23.8.1 SW protocol introduction

This synchronous serial protocol uses two pins:

- SWCLK: clock from host to target
- SWDIO: bidirectional

The protocol allows two banks of registers (DPACC registers and APACC registers) to be read and written to.

Bits are transferred LSB-first on the wire.

For SWDIO bidirectional management, the line must be pulled-up on the board (100 K $\Omega$  recommended by Arm<sup>®</sup>).

Each time the direction of SWDIO changes in the protocol, a turnaround time is inserted where the line is not driven by the host nor the target. By default, this turnaround time is one bit time, however this can be adjusted by configuring the SWCLK frequency.

### 23.8.2 SW protocol sequence

Each sequence consist of three phases:

1. Packet request (8 bits) transmitted by the host
2. Acknowledge response (3 bits) transmitted by the target
3. Data transfer phase (33 bits) transmitted by the host or the target

**Table 139. Packet request (8-bits)**

| Bit | Name   | Description                                                                      |
|-----|--------|----------------------------------------------------------------------------------|
| 0   | Start  | Must be "1"                                                                      |
| 1   | APnDP  | 0: DP Access<br>1: AP Access                                                     |
| 2   | RnW    | 0: Write Request<br>1: Read Request                                              |
| 4:3 | A[3:2] | Address field of the DP or AP registers (refer to <a href="#">Table 138</a> )    |
| 5   | Parity | Single bit parity of preceding bits                                              |
| 6   | Stop   | 0                                                                                |
| 7   | Park   | Not driven by the host. Must be read as "1" by the target because of the pull-up |

Refer to the Cortex<sup>®</sup>-M4 with FPU r0p1 *TRM* for a detailed description of DPACC and APACC registers.

The packet request is always followed by the turnaround time (default 1 bit) where neither the host nor target drive the line.

Table 140. ACK response (3 bits)

| Bit  | Name | Description                        |
|------|------|------------------------------------|
| 0..2 | ACK  | 001: FAULT<br>010: WAIT<br>100: OK |

The ACK Response must be followed by a turnaround time only if it is a READ transaction or if a WAIT or FAULT acknowledge has been received.

Table 141. DATA transfer (33 bits)

| Bit   | Name           | Description                       |
|-------|----------------|-----------------------------------|
| 0..31 | WDATA or RDATA | Write or Read data                |
| 32    | Parity         | Single parity of the 32 data bits |

The DATA transfer must be followed by a turnaround time only if it is a READ transaction.

### 23.8.3 SW-DP state machine (reset, idle states, ID code)

The State Machine of the SW-DP has an internal ID code which identifies the SW-DP. It follows the JEP-106 standard. This ID code is the default Arm® one and is set to **0x2BA01477** (corresponding to Cortex®-M4 with FPU r0p1).

*Note:* Note that the SW-DP state machine is inactive until the target reads this ID code.

- The SW-DP state machine is in RESET STATE either after power-on reset, or after the DP has switched from JTAG to SWD or after the line is high for more than 50 cycles
- The SW-DP state machine is in IDLE STATE if the line is low for at least two cycles after RESET state.
- After RESET state, it is **mandatory** to first enter into an IDLE state AND to perform a READ access of the DP-SW ID CODE register. Otherwise, the target will issue a FAULT acknowledge response on another transactions.

Further details of the SW-DP state machine can be found in the Cortex®-M4 with FPU r0p1 *TRM* and the *CoreSight Design Kit r0p1 TRM*.

### 23.8.4 DP and AP read/write accesses

- Read accesses to the DP are not posted: the target response can be immediate (if ACK=OK) or can be delayed (if ACK=WAIT).
- Read accesses to the AP are posted. This means that the result of the access is returned on the next transfer. If the next access to be done is NOT an AP access, then the DP-RDBUFF register must be read to obtain the result.  
The READOK flag of the DP-CTRL/STAT register is updated on every AP read access or RDBUFF read request to know if the AP read access was successful.
- The SW-DP implements a write buffer (for both DP or AP writes), that enables it to accept a write operation even when other transactions are still outstanding. If the write buffer is full, the target acknowledge response is "WAIT". With the exception of

IDCODE read or CTRL/STAT read or ABORT write which are accepted even if the write buffer is full.

- Because of the asynchronous clock domains SWCLK and HCLK, two extra SWCLK cycles are needed after a write transaction (after the parity bit) to make the write effective internally. These cycles should be applied while driving the line low (IDLE state)  
This is particularly important when writing the CTRL/STAT for a power-up request. If the next transaction (requiring a power-up) occurs immediately, it will fail.

### 23.8.5 SW-DP registers

Access to these registers are initiated when APnDP=0

**Table 142. SW-DP registers**

| A[3:2] | R/W        | CTRLSEL bit of SELECT register | Register     | Notes                                                                                                                                                                                                                                                                |
|--------|------------|--------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00     | Read       | -                              | IDCODE       | The manufacturer code is not set to ST code. <b>0x2BA01477</b> (identifies the SW-DP)                                                                                                                                                                                |
| 00     | Write      | -                              | ABORT        | -                                                                                                                                                                                                                                                                    |
| 01     | Read/Write | 0                              | DP-CTRL/STAT | Purpose is to:<br>– request a system or debug power-up<br>– configure the transfer operation for AP accesses<br>– control the pushed compare and pushed verify operations.<br>– read some status flags (overrun, power-up acknowledges)                              |
| 01     | Read/Write | 1                              | WIRE CONTROL | Purpose is to configure the physical serial port protocol (like the duration of the turnaround time)                                                                                                                                                                 |
| 10     | Read       | -                              | READ RESEND  | Enables recovery of the read data from a corrupted debugger transfer, without repeating the original AP transfer.                                                                                                                                                    |
| 10     | Write      | -                              | SELECT       | The purpose is to select the current access port and the active 4-words register window                                                                                                                                                                              |
| 11     | Read/Write | -                              | READ BUFFER  | This read buffer is useful because AP accesses are posted (the result of a read AP request is available on the next AP transaction).<br>This read buffer captures data from the AP, presented as the result of a previous read, without initiating a new transaction |

### 23.8.6 SW-AP registers

Access to these registers are initiated when APnDP=1

There are many AP Registers (see AHB-AP) addressed as the combination of:

- The shifted value A[3:2]
- The current value of the DP SELECT register

## 23.9 AHB-AP (AHB access port) - valid for both JTAG-DP and SW-DP

### Features:

- System access is independent of the processor status.
- Either SW-DP or JTAG-DP accesses AHB-AP.
- The AHB-AP is an AHB master into the Bus Matrix. Consequently, it can access all the data buses (Dcode Bus, System Bus, internal and external PPB bus) but the ICode bus.
- Bitband transactions are supported.
- AHB-AP transactions bypass the FPB.

The address of the 32-bits AHB-AP registers are 6-bits wide (up to 64 words or 256 bytes) and consists of:

- c) Bits [7:4] = the bits [7:4] APBANKSEL of the DP SELECT register
- d) Bits [3:2] = the 2 address bits of A[3:2] of the 35-bit packet request for SW-DP.

The AHB-AP of the Cortex<sup>®</sup>-M4 with FPU includes 9 x 32-bits registers:

**Table 143. Cortex<sup>®</sup>-M4 with FPU AHB-AP registers**

| Address offset | Register name                  | Notes                                                                                                                         |
|----------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 0x00           | AHB-AP Control and Status Word | Configures and controls transfers through the AHB interface (size, hprot, status on current transfer, address increment type) |
| 0x04           | AHB-AP Transfer Address        | -                                                                                                                             |
| 0x0C           | AHB-AP Data Read/Write         | -                                                                                                                             |
| 0x10           | AHB-AP Banked Data 0           | Directly maps the 4 aligned data words without rewriting the Transfer Address Register.                                       |
| 0x14           | AHB-AP Banked Data 1           |                                                                                                                               |
| 0x18           | AHB-AP Banked Data 2           |                                                                                                                               |
| 0x1C           | AHB-AP Banked Data 3           |                                                                                                                               |
| 0xF8           | AHB-AP Debug ROM Address       | Base Address of the debug interface                                                                                           |
| 0xFC           | AHB-AP ID Register             | -                                                                                                                             |

Refer to the Cortex<sup>®</sup>-M4 with FPU *r0p1 TRM* for further details.

## 23.10 Core debug

Core debug is accessed through the core debug registers. Debug access to these registers is by means of the *Advanced High-performance Bus* (AHB-AP) port. The processor can access these registers directly over the internal *Private Peripheral Bus* (PPB).

It consists of 4 registers:

**Table 144. Core debug registers**

| Register | Description                                                                                                                                                                                            |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DHCSR    | The 32-bit Debug Halting Control and Status Register<br>This provides status information about the state of the processor enable core debug halt and step the processor                                |
| DCRSR    | The 17-bit Debug Core Register Selector Register:<br>This selects the processor register to transfer data to or from.                                                                                  |
| DCRDR    | The 32-bit Debug Core Register Data Register:<br>This holds data for reading and writing registers to and from the processor selected by the DCRSR (Selector) register.                                |
| DEMCR    | The 32-bit Debug Exception and Monitor Control Register:<br>This provides Vector Catching and Debug Monitor Control. This register contains a bit named <b>TRCENA</b> which enable the use of a TRACE. |

*Note:* **Important:** these registers are not reset by a system reset. They are only reset by a power-on reset.

Refer to the Cortex®-M4 with FPU r0p1 TRM for further details.

To Halt on reset, it is necessary to:

- enable the bit0 (VC\_CORRESET) of the Debug and Exception Monitor Control Register
- enable the bit0 (C\_DEBUGEN) of the Debug Halting Control and Status Register.

## 23.11 Capability of the debugger host to connect under system reset

The reset system of the STM32F411xC/E MCU comprises the following reset sources:

- POR (power-on reset) which asserts a RESET at each power-up.
- Internal watchdog reset
- Software reset
- External reset

The Cortex<sup>®</sup>-M4 with FPU differentiates the reset of the debug part (generally PORRESETn) and the other one (SYSRESETn)

This way, it is possible for the debugger to connect under System Reset, programming the Core Debug Registers to halt the core when fetching the reset vector. Then the host can release the system reset and the core will immediately halt without having executed any instructions. In addition, it is possible to program any debug features under System Reset.

*Note: It is highly recommended for the debugger host to connect (set a breakpoint in the reset vector) under system reset.*

## 23.12 FPB (Flash patch breakpoint)

The FPB unit:

- implements hardware breakpoints
- patches code and data from code space to system space. This feature gives the possibility to correct software bugs located in the Code Memory Space.

The use of a Software Patch or a Hardware Breakpoint is exclusive.

The FPB consists of:

- 2 literal comparators for matching against literal loads from Code Space and remapping to a corresponding area in the System Space.
- 6 instruction comparators for matching against instruction fetches from Code Space. They can be used either to remap to a corresponding area in the System Space or to generate a Breakpoint Instruction to the core.

## 23.13 DWT (data watchpoint trigger)

The DWT unit consists of four comparators. They are configurable as:

- a hardware watchpoint or
- a trigger to an ETM or
- a PC sampler or
- a data address sampler

The DWT also provides some means to give some profiling informations. For this, some counters are accessible to give the number of:

- Clock cycle
- Folded instructions
- Load store unit (LSU) operations
- Sleep cycles
- CPI (clock per instructions)
- Interrupt overhead

## 23.14 ITM (instrumentation trace macrocell)

### 23.14.1 General description

The ITM is an application-driven trace source that supports *printf* style debugging to trace *Operating System* (OS) and application events, and emits diagnostic system information. The ITM emits trace information as packets which can be generated as:

- **Software trace.** Software can write directly to the ITM stimulus registers to emit packets.
- **Hardware trace.** The DWT generates these packets, and the ITM emits them.
- **Time stamping.** Timestamps are emitted relative to packets. The ITM contains a 21-bit counter to generate the timestamp. The Cortex<sup>®</sup>-M4 with FPU clock or the bit clock rate of the *Serial Wire Viewer* (SWV) output clocks the counter.

The packets emitted by the ITM are output to the TPIU (Trace Port Interface Unit). The formatter of the TPIU adds some extra packets (refer to TPIU) and then output the complete packets sequence to the debugger host.

The bit TRCEN of the Debug Exception and Monitor Control Register must be enabled before programming or using the ITM.

### 23.14.2 Time stamp packets, synchronization and overflow packets

Time stamp packets encode time stamp information, generic control and synchronization. It uses a 21-bit timestamp counter (with possible prescalers) which is reset at each time stamp packet emission. This counter can be either clocked by the CPU clock or the SWV clock.

A synchronization packet consists of 6 bytes equal to 0x80\_00\_00\_00\_00\_00 which is emitted to the TPIU as 00 00 00 00 00 80 (LSB emitted first).

A synchronization packet is a timestamp packet control. It is emitted at each DWT trigger.

For this, the DWT must be configured to trigger the ITM: the bit CYCCNTENA (bit0) of the DWT Control Register must be set. In addition, the bit2 (SYNCENA) of the ITM Trace Control Register must be set.

*Note: If the SYNENA bit is not set, the DWT generates Synchronization triggers to the TPIU which will send only TPIU synchronization packets and not ITM synchronization packets.*

An overflow packet consists is a special timestamp packets which indicates that data has been written but the FIFO was full.

**Table 145. Main ITM registers**

| Address                | Register                     | Details                                                                                                                                                   |
|------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| @E0000FB0              | ITM lock access              | Write 0xC5ACCE55 to unlock Write Access to the other ITM registers                                                                                        |
| @E0000E80              | ITM trace control            | Bits 31-24 = Always 0                                                                                                                                     |
|                        |                              | Bits 23 = Busy                                                                                                                                            |
|                        |                              | Bits 22-16 = 7-bits ATB ID which identifies the source of the trace data.                                                                                 |
|                        |                              | Bits 15-10 = Always 0                                                                                                                                     |
|                        |                              | Bits 9:8 = TSPrescale = Time Stamp Prescaler                                                                                                              |
|                        |                              | Bits 7-5 = Reserved                                                                                                                                       |
|                        |                              | Bit 4 = SWOENA = Enable SWV behavior (to clock the timestamp counter by the SWV clock).                                                                   |
|                        |                              | Bit 3 = DWTENA: Enable the DWT Stimulus                                                                                                                   |
|                        |                              | Bit 2 = SYNCENA: this bit must be to 1 to enable the DWT to generate synchronization triggers so that the TPIU can then emit the synchronization packets. |
|                        |                              | Bit 1 = TSENA (Timestamp Enable)                                                                                                                          |
|                        |                              | Bit 0 = ITMENA: Global Enable Bit of the ITM                                                                                                              |
| @E0000E40              | ITM trace privilege          | Bit 3: mask to enable tracing ports31:24                                                                                                                  |
|                        |                              | Bit 2: mask to enable tracing ports23:16                                                                                                                  |
|                        |                              | Bit 1: mask to enable tracing ports15:8                                                                                                                   |
|                        |                              | Bit 0: mask to enable tracing ports7:0                                                                                                                    |
| @E0000E00              | ITM trace enable             | Each bit enables the corresponding Stimulus port to generate trace.                                                                                       |
| @E0000000-<br>E000007C | Stimulus port registers 0-31 | Write the 32-bits data on the selected Stimulus Port (32 available) to be traced out.                                                                     |

### Example of configuration

To output a simple value to the TPIU:

- Configure the TPIU and assign TRACE I/Os by configuring the DBGMCU\_CR (refer to [Section 23.17.2: TRACE pin assignment](#) and [Section 23.16.3: Debug MCU configuration register](#))
- Write 0xC5ACCE55 to the ITM Lock Access Register to unlock the write access to the ITM registers
- Write 0x00010005 to the ITM Trace Control Register to enable the ITM with Sync enabled and an ATB ID different from 0x00
- Write 0x1 to the ITM Trace Enable Register to enable the Stimulus Port 0
- Write 0x1 to the ITM Trace Privilege Register to unmask stimulus ports 7:0
- Write the value to output in the Stimulus Port Register 0: this can be done by software (using a printf function)

## 23.15 ETM (Embedded trace macrocell)

### 23.15.1 ETM general description

The ETM enables the reconstruction of program execution. Data are traced using the Data Watchpoint and Trace (DWT) component or the Instruction Trace Macrocell (ITM) whereas instructions are traced using the Embedded Trace Macrocell (ETM).

The ETM transmits information as packets and is triggered by embedded resources. These resources must be programmed independently and the trigger source is selected using the Trigger Event Register (0xE0041008). An event could be a simple event (address match from an address comparator) or a logic equation between 2 events. The trigger source is one of the fourth comparators of the DWT module, The following events can be monitored:

- Clock cycle matching
- Data address matching

For more informations on the trigger resources refer to [Section 23.13: DWT \(data watchpoint trigger\)](#).

The packets transmitted by the ETM are output to the TPIU (Trace Port Interface Unit). The formatter of the TPIU adds some extra packets (refer to [Section 23.17: TPIU \(trace port interface unit\)](#)) and then outputs the complete packet sequence to the debugger host.

### 23.15.2 ETM signal protocol and packet types

This part is described in the chapter 7 ETMv3 Signal Protocol of the Arm® IHI 0014N document.

### 23.15.3 Main ETM registers

For more information on registers refer to the chapter 3 of the Arm® IHI 0014N specification.

**Table 146. Main ETM registers**

| Address    | Register                 | Details                                                                                                |
|------------|--------------------------|--------------------------------------------------------------------------------------------------------|
| 0xE0041FB0 | ETM Lock Access          | Write 0xC5ACCE55 to unlock the write access to the other ETM registers.                                |
| 0xE0041000 | ETM Control              | This register controls the general operation of the ETM, for instance how tracing is enabled.          |
| 0xE0041010 | ETM Status               | This register provides information about the current status of the trace and trigger logic.            |
| 0xE0041008 | ETM Trigger Event        | This register defines the event that will control trigger.                                             |
| 0xE004101C | ETM Trace Enable Control | This register defines which comparator is selected.                                                    |
| 0xE0041020 | ETM Trace Enable Event   | This register defines the trace enabling event.                                                        |
| 0xE0041024 | ETM Trace Start/Stop     | This register defines the traces used by the trigger source to start and stop the trace, respectively. |

### 23.15.4 ETM configuration example

To output a simple value to the TPIU:

- Configure the TPIU and enable the I/O\_TRACEN to assign TRACE I/Os in the STM32F411xC/E debug configuration register.
- Write 0xC5AC CE55 to the ETM Lock Access Register to unlock the write access to the ITM registers
- Write 0x0000 1D1E to the ETM control register (configure the trace)
- Write 0x0000 406F to the ETM Trigger Event register (define the trigger event)
- Write 0x0000 006F to the ETM Trace Enable Event register (define an event to start/stop)
- Write 0x0200 0000 to the ETM Trace Start/stop register (enable the trace)
- Write 0x0000191E to the ETM Control Register (end of configuration)

## 23.16 MCU debug component (DBGMCU)

The MCU debug component helps the debugger provide support for:

- Low-power modes
- Clock control for timers, watchdog and I2C during a breakpoint
- Control of the trace pins assignment

### 23.16.1 Debug support for low-power modes

To enter low-power mode, the instruction WFI or WFE must be executed.

The MCU implements several low-power modes which can either deactivate the CPU clock or reduce the power of the CPU.

The core does not allow FCLK or HCLK to be turned off during a debug session. As these are required for the debugger connection, during a debug, they must remain active. The MCU integrates special means to allow the user to debug software in low-power modes.

For this, the debugger host must first set some debug configuration registers to change the low-power mode behavior:

- In Sleep mode, DBG\_SLEEP bit of DBGMCU\_CR register must be previously set by the debugger. This will feed HCLK with the same clock that is provided to FCLK (system clock previously configured by the software).
- In Stop mode, the bit DBG\_STOP must be previously set by the debugger. This will enable the internal RC oscillator clock to feed FCLK and HCLK in STOP mode.

### 23.16.2 Debug support for timers, watchdog and I<sup>2</sup>C

During a breakpoint, it is necessary to choose how the counter of timers and watchdog should behave:

- They can continue to count inside a breakpoint. This is usually required when a PWM is controlling a motor, for example.
- They can stop to count inside a breakpoint. This is required for watchdog purposes.

For the I<sup>2</sup>C, the user can choose to block the SMBUS timeout during a breakpoint.

For timers having complementary outputs, when the counter is stopped (DBG\_TIMx\_STOP = 1), the outputs are disabled (as if the MOE bit was reset) for safety purposes.

### 23.16.3 Debug MCU configuration register

This register allows the configuration of the MCU under DEBUG. This concerns:

- Low-power mode support
- Timer and watchdog counter support
- Trace pin assignment

This DBGMCU\_CR is mapped on the External PPB bus at address 0xE0042004

It is asynchronously reset by the PORESET (and not the system reset). It can be written by the debugger under system reset.

If the debugger host does not support these features, it is still possible for the user software to write to these registers.

#### DBGMCU\_CR register

Address: 0xE004 2004

Only 32-bit access supported

POR Reset: 0x0000 0000 (not reset by system reset)

|          |    |    |    |    |    |    |    |                  |    |            |          |    |             |          |           |
|----------|----|----|----|----|----|----|----|------------------|----|------------|----------|----|-------------|----------|-----------|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23               | 22 | 21         | 20       | 19 | 18          | 17       | 16        |
| Reserved |    |    |    |    |    |    |    |                  |    |            |          |    |             |          |           |
| 15       | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7                | 6  | 5          | 4        | 3  | 2           | 1        | 0         |
| Reserved |    |    |    |    |    |    |    | TRACE_MODE [1:0] |    | TRACE_IOEN | Reserved |    | DBG_STANDBY | DBG_STOP | DBG_SLEEP |
|          |    |    |    |    |    |    |    | rw               | rw | rw         |          |    | rw          | rw       | rw        |

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:5 **TRACE\_MODE[1:0] and TRACE\_IOEN**: Trace pin assignment control

- With TRACE\_IOEN=0:
  - TRACE\_MODE=xx: TRACE pins not assigned (default state)
- With TRACE\_IOEN=1:
  - TRACE\_MODE=00: TRACE pin assignment for Asynchronous Mode
  - TRACE\_MODE=01: TRACE pin assignment for Synchronous Mode with a TRACEDATA size of 1
  - TRACE\_MODE=10: TRACE pin assignment for Synchronous Mode with a TRACEDATA size of 2
  - TRACE\_MODE=11: TRACE pin assignment for Synchronous Mode with a TRACEDATA size of 4

Bits 4:3 Reserved, must be kept at reset value.

Bit 2 **DBG\_STANDBY**: Debug Standby mode

0: (FCLK=Off, HCLK=Off) The whole digital part is unpowered.

From software point of view, exiting from Standby is identical than fetching reset vector (except a few status bit indicated that the MCU is resuming from Standby)

1: (FCLK=On, HCLK=On) In this case, the digital part is not unpowered and FCLK and HCLK are provided by the internal RC oscillator which remains active. In addition, the MCU generate a system reset during Standby mode so that exiting from Standby is identical than fetching from reset

Bit 1 **DBG\_STOP**: Debug Stop mode

0: (FCLK=Off, HCLK=Off) In STOP mode, the clock controller disables all clocks (including HCLK and FCLK). When exiting from STOP mode, the clock configuration is identical to the one after RESET (CPU clocked by the 8 MHz internal RC oscillator (HSI)). Consequently, the software must reprogram the clock controller to enable the PLL, the Xtal, etc.

1: (FCLK=On, HCLK=On) In this case, when entering STOP mode, FCLK and HCLK are provided by the internal RC oscillator which remains active in STOP mode. When exiting STOP mode, the software must reprogram the clock controller to enable the PLL, the Xtal, etc. (in the same way it would do in case of DBG\_STOP=0)

Bit 0 **DBG\_SLEEP**: Debug Sleep mode

0: (FCLK=On, HCLK=Off) In Sleep mode, FCLK is clocked by the system clock as previously configured by the software while HCLK is disabled.

In Sleep mode, the clock controller configuration is not reset and remains in the previously programmed state. Consequently, when exiting from Sleep mode, the software does not need to reconfigure the clock controller.

1: (FCLK=On, HCLK=On) In this case, when entering Sleep mode, HCLK is fed by the same clock that is provided to FCLK (system clock as previously configured by the software).

### 23.16.4 Debug MCU APB1 freeze register (DBGMCU\_APB1\_FZ)

The DBGMCU\_APB1\_FZ register is used to configure the MCU under Debug. It concerns APB1 peripherals. It is mapped on the external PPB bus at address 0xE004 2008.

The register is asynchronously reset by the POR (and not the system reset). It can be written by the debugger under system reset.

Address : 0xE004 2008

Only 32-bits access are supported.

Power-on reset (POR): 0x0000 0000 (not reset by system reset)

|          |    |    |    |               |               |              |          |                        |                        |                        |          |               |               |               |               |
|----------|----|----|----|---------------|---------------|--------------|----------|------------------------|------------------------|------------------------|----------|---------------|---------------|---------------|---------------|
| 31       | 30 | 29 | 28 | 27            | 26            | 25           | 24       | 23                     | 22                     | 21                     | 20       | 19            | 18            | 17            | 16            |
| Reserved |    |    |    |               |               |              |          | DBG_I2C3_SMBUS_TIMEOUT | DBG_I2C2_SMBUS_TIMEOUT | DBG_I2C1_SMBUS_TIMEOUT | Reserved |               |               |               |               |
|          |    |    |    |               |               |              |          | rw                     | rw                     | rw                     |          |               |               |               |               |
| 15       | 14 | 13 | 12 | 11            | 10            | 9            | 8        | 7                      | 6                      | 5                      | 4        | 3             | 2             | 1             | 0             |
| Reserved |    |    |    | DBG_IWDG_STOP | DBG_WWDG_STOP | DBG_RTC_STOP | Reserved |                        |                        |                        |          | DBG_TIM5_STOP | DBG_TIM4_STOP | DBG_TIM3_STOP | DBG_TIM2_STOP |
|          |    |    |    |               |               |              |          |                        |                        |                        |          |               |               |               |               |

Bits 31:24 Reserved, must be kept at reset value.

Bit 23 **DBG\_I2C3\_SMBUS\_TIMEOUT**: SMBUS timeout mode stopped when Core is halted

- 0: Same behavior as in normal mode
- 1: The SMBUS timeout is frozen

Bit 22 **DBG\_I2C2\_SMBUS\_TIMEOUT**: SMBUS timeout mode stopped when Core is halted

- 0: Same behavior as in normal mode
- 1: The SMBUS timeout is frozen

Bit 21 **DBG\_I2C1\_SMBUS\_TIMEOUT**: SMBUS timeout mode stopped when Core is halted

- 0: Same behavior as in normal mode
- 1: The SMBUS timeout is frozen

Bit 20:13 Reserved, must be kept at reset value.

Bit 12 **DBG\_IWDG\_STOP**: Debug independent watchdog stopped when core is halted

- 0: The independent watchdog counter clock continues even if the core is halted
- 1: The independent watchdog counter clock is stopped when the core is halted

Bit 11 **DBG\_WWDG\_STOP**: Debug Window Watchdog stopped when Core is halted

- 0: The window watchdog counter clock continues even if the core is halted
- 1: The window watchdog counter clock is stopped when the core is halted

Bit 10 **DBG\_RTC\_STOP**: RTC stopped when Core is halted

0: The RTC counter clock continues even if the core is halted

1: The RTC counter clock is stopped when the core is halted

Bit 9:4 Reserved, must be kept at reset value.

Bits 3:0 **DBG\_TIMx\_STOP**: TIMx counter stopped when core is halted (x=2..5)

0: The clock of the involved timer counter is fed even if the core is halted

1: The clock of the involved timer counter is stopped and the outputs are disabled when the core is halted

### 23.16.5 Debug MCU APB2 Freeze register (DBGMCU\_APB2\_FZ)

The DBGMCU\_APB2\_FZ register is used to configure the MCU under Debug. It concerns APB2 peripherals.

This register is mapped on the external PPB bus at address 0xE004 200C

It is asynchronously reset by the POR (and not the system reset). It can be written by the debugger under system reset.

Address: 0xE004 200C

Only 32-bit access is supported.

POR: 0x0000 0000 (not reset by system reset)

|          |    |    |    |    |    |    |    |    |    |    |    |    |                |                |               |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----------------|----------------|---------------|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18             | 17             | 16            |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    | DBG_TIM11_STOP | DBG_TIM10_STOP | DBG_TIM9_STOP |
|          |    |    |    |    |    |    |    |    |    |    |    |    | rw             | rw             | rw            |
| 15       | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2              | 1              | 0             |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |                | DBG_TIM1_STOP  |               |
|          |    |    |    |    |    |    |    |    |    |    |    |    |                | rw             |               |

Bits 31:19 Reserved, must be kept at reset value.

Bits 18:16 **DBG\_TIMx\_STOP**: TIMx counter stopped when core is halted (x=9..11)

0: The clock of the involved timer counter is fed even if the core is halted

1: The clock of the involved timer counter is stopped and the outputs are disabled when the core is halted

Bits 15:1 Reserved, must be kept at reset value.

Bit 0 **DBG\_TIM1\_STOP**: TIM1 counter stopped when core is halted

0: The clock of the involved timer counter is fed even if the core is halted

1: The clock of the involved timer counter is stopped and the outputs are disabled when the core is halted

## 23.17 TPIU (trace port interface unit)

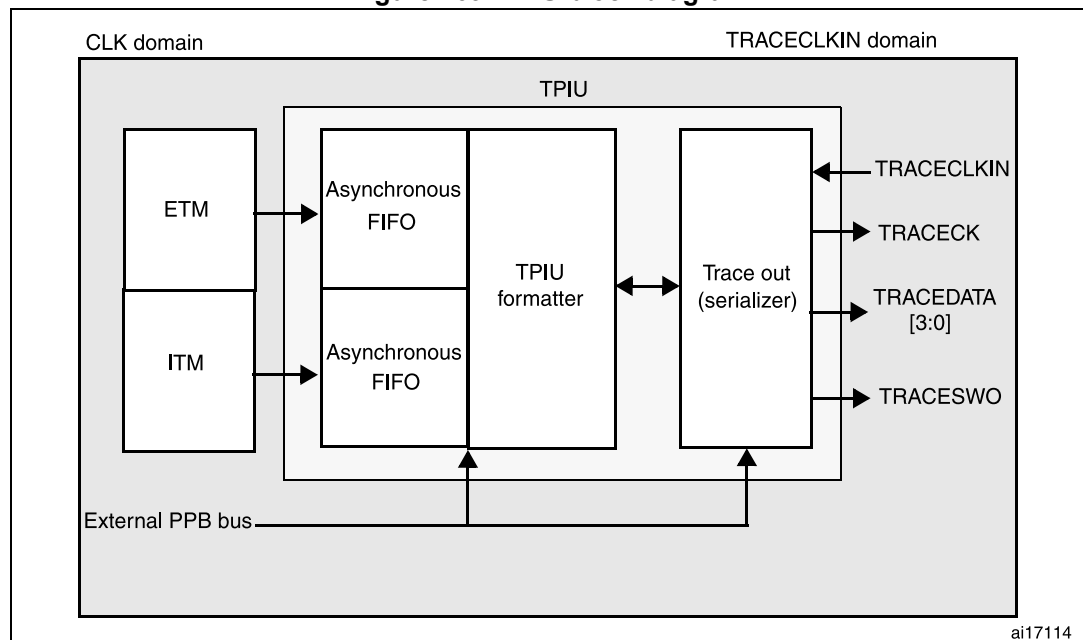
### 23.17.1 Introduction

The TPIU acts as a bridge between the on-chip trace data from the ITM and the ETM.

The output data stream encapsulates the trace source ID, that is then captured by a *trace port analyzer* (TPA).

The core embeds a simple TPIU, especially designed for low-cost debug (consisting of a special version of the CoreSight TPIU).

**Figure 269. TPIU block diagram**



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## 23.17.2 TRACE pin assignment

- **Asynchronous mode**  
The asynchronous mode requires 1 extra pin and is available on all packages. It is only available if using Serial Wire mode (not in JTAG mode).

**Table 147. Asynchronous TRACE pin assignment**

| TPUI pin name | Trace synchronous mode |                         | pin assignment |
|---------------|------------------------|-------------------------|----------------|
|               | Type                   | Description             |                |
| TRACESWO      | O                      | TRACE Async Data Output | PB3            |

- **Synchronous mode**  
The synchronous mode requires from 2 to 6 extra pins depending on the data trace size and is only available in the larger packages. In addition it is available in JTAG mode and in Serial Wire mode and provides better bandwidth output capabilities than asynchronous trace.

**Table 148. Synchronous TRACE pin assignment**

| TPUI pin name | Trace synchronous mode |                                              | STM32F411xC/E pin assignment |
|---------------|------------------------|----------------------------------------------|------------------------------|
|               | Type                   | Description                                  |                              |
| TRACECK       | O                      | TRACE Clock                                  | PE2                          |
| TRACED[3:0]   | O                      | TRACE Sync Data Outputs<br>Can be 1, 2 or 4. | PE[6:3]                      |

### TPUI TRACE pin assignment

By default, these pins are NOT assigned. They can be assigned by setting the TRACE\_IOEN and TRACE\_MODE bits in the **MCU Debug component configuration register**. This configuration has to be done by the debugger host.

In addition, the number of pins to assign depends on the trace configuration (asynchronous or synchronous).

- **Asynchronous mode:** 1 extra pin is needed
- **Synchronous mode:** from 2 to 5 extra pins are needed depending on the size of the data trace port register (1, 2 or 4):
  - TRACECK
  - TRACED(0) if port size is configured to 1, 2 or 4
  - TRACED(1) if port size is configured to 2 or 4
  - TRACED(2) if port size is configured to 4
  - TRACED(3) if port size is configured to 4

To assign the TRACE pin, the debugger host must program the bits TRACE\_IOEN and TRACE\_MODE[1:0] of the Debug MCU configuration Register (DBGMCU\_CR). By default the TRACE pins are not assigned.

This register is mapped on the external PPB and is reset by the PORESET (and not by the SYSTEM reset). It can be written by the debugger under SYSTEM reset.

Table 149. Flexible TRACE pin assignment

| DBGMCU_CR register |                  | Pins assigned for:       | TRACE IO pin assigned   |              |                 |                           |                 |                 |
|--------------------|------------------|--------------------------|-------------------------|--------------|-----------------|---------------------------|-----------------|-----------------|
| TRACE_IOEN         | TRACE_MODE [1:0] |                          | PB3 /JTDO/ TRACESWO     | PE2/ TRACECK | PE3 / TRACED[0] | PE4 / TRACED[1]           | PE5 / TRACED[2] | PE6 / TRACED[3] |
| 0                  | XX               | No Trace (default state) | Released <sup>(1)</sup> | -            |                 |                           |                 |                 |
| 1                  | 00               | Asynchronous Trace       | TRACESWO                | -            | -               | Released (usable as GPIO) |                 |                 |
| 1                  | 01               | Synchronous Trace 1 bit  | Released <sup>(1)</sup> | TRACECK      | TRACED[0]       | -                         | -               | -               |
| 1                  | 10               | Synchronous Trace 2 bit  |                         | TRACECK      | TRACED[0]       | TRACED[1]                 | -               | -               |
| 1                  | 11               | Synchronous Trace 4 bit  |                         | TRACECK      | TRACED[0]       | TRACED[1]                 | TRACED[2]       | TRACED[3]       |

1. When Serial Wire mode is used, it is released. But when JTAG is used, it is assigned to JTDO.

**Note:** By default, the TRACECLKIN input clock of the TPIU is tied to GND. It is assigned to HCLK two clock cycles after the bit TRACE\_IOEN has been set.

The debugger must then program the Trace Mode by writing the PROTOCOL[1:0] bits in the SPP\_R (Selected Pin Protocol) register of the TPIU.

- PROTOCOL=00: Trace Port Mode (synchronous)
- PROTOCOL=01 or 10: Serial Wire (Manchester or NRZ) Mode (asynchronous mode). Default state is 01

It then also configures the TRACE port size by writing the bits [3:0] in the CPSPS\_R (Current Sync Port Size Register) of the TPIU:

- 0x1 for 1 pin (default state)
- 0x2 for 2 pins
- 0x8 for 4 pins

### 23.17.3 TPUI formatter

The formatter protocol outputs data in 16-byte frames:

- seven bytes of data
- eight bytes of mixed-use bytes consisting of:
  - 1 bit (LSB) to indicate it is a DATA byte ('0') or an ID byte ('1').
  - 7 bits (MSB) which can be data or change of source ID trace.
- one byte of auxiliary bits where each bit corresponds to one of the eight mixed-use bytes:
  - if the corresponding byte was a data, this bit gives bit0 of the data.
  - if the corresponding byte was an ID change, this bit indicates when that ID change takes effect.

*Note:* Refer to the Arm® CoreSight Architecture Specification v1.0 (Arm® IHI 0029B) for further information

### 23.17.4 TPUI frame synchronization packets

The TPUI can generate two types of synchronization packets:

- The Frame Synchronization packet (or Full Word Synchronization packet)  
It consists of the word: 0x7F\_FF\_FF\_FF (LSB emitted first). This sequence can not occur at any other time provided that the ID source code 0x7F has not been used.  
It is output periodically **between** frames.  
In continuous mode, the TPA must discard all these frames once a synchronization frame has been found.
- The Half-Word Synchronization packet  
It consists of the half word: 0x7F\_FF (LSB emitted first).  
It is output periodically **between or within** frames.  
These packets are only generated in continuous mode and enable the TPA to detect that the TRACE port is in IDLE mode (no TRACE to be captured). When detected by the TPA, it must be discarded.

### 23.17.5 Transmission of the synchronization frame packet

There is no Synchronization Counter register implemented in the TPIU of the core. Consequently, the synchronization trigger can only be generated by the **DWT**. Refer to the registers DWT Control Register (bits SYNCTAP[11:10]) and the DWT Current PC Sampler Cycle Count Register.

The TPUI Frame synchronization packet (0x7F\_FF\_FF\_FF) is emitted:

- after each TPIU reset release. This reset is synchronously released with the rising edge of the TRACECLKIN clock. This means that this packet is transmitted when the TRACE\_IOEN bit in the DBGMCU\_CFG register is set. In this case, the word 0x7F\_FF\_FF\_FF is not followed by any formatted packet.
- at each DWT trigger (assuming DWT has been previously configured). Two cases occur:
  - If the bit SYNENA of the ITM is reset, only the word 0x7F\_FF\_FF\_FF is emitted without any formatted stream which follows.
  - If the bit SYNENA of the ITM is set, then the ITM synchronization packets will follow (0x80\_00\_00\_00\_00\_00), formatted by the TPUI (trace source ID added).

### 23.17.6 Synchronous mode

The trace data output size can be configured to 4, 2 or 1 pin: TRACED(3:0)

The output clock is output to the debugger (TRACECK)

Here, TRACECLKIN is driven internally and is connected to HCLK only when TRACE is used.

*Note:* In this synchronous mode, it is not required to provide a stable clock frequency.

The TRACE I/Os (including TRACECK) are driven by the rising edge of TRACCLKIN (equal to HCLK). Consequently, the output frequency of TRACECK is equal to HCLK/2.

### 23.17.7 Asynchronous mode

This is a low cost alternative to output the trace using only 1 pin: this is the asynchronous output pin TRACESWO. Obviously there is a limited bandwidth.

TRACESWO is multiplexed with JTDO when using the SW-DP pin. This way, this functionality is available in all STM32F411xC/E packages.

This asynchronous mode requires a constant frequency for TRACECLKIN. For the standard UART (NRZ) capture mechanism, 5% accuracy is needed. The Manchester encoded version is tolerant up to 10%.

### 23.17.8 TRACECLKIN connection inside the STM32F411xC/E

In the STM32F411xC/E, this TRACECLKIN input is internally connected to HCLK. This means that when in asynchronous trace mode, the application is restricted to use to time frames where the CPU frequency is stable.

**Note:** ***Important:** when using asynchronous trace: it is important to be aware that:*

*The default clock of the STM32F411xC/E MCUs is the internal RC oscillator. Its frequency under reset is different from the one after reset release. This is because the RC calibration is the default one under system reset and is updated at each system reset release.*

*Consequently, the trace port analyzer (TPA) should not enable the trace (with the TRACE\_IOEN bit) under system reset, because a Synchronization Frame Packet will be issued with a different bit time than trace packets which will be transmitted after reset release.*

### 23.17.9 TPIU registers

The TPIU APB registers can be read and written only if the bit TRCENA of the Debug Exception and Monitor Control Register (DEMCR) is set. Otherwise, the registers are read as zero (the output of this bit enables the PCLK of the TPIU).

**Table 150. Important TPIU registers**

| Address    | Register              | Description                                                                                                                                                                                                                             |
|------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0xE0040004 | Current port size     | Allows the trace port size to be selected:<br>Bit 0: Port size = 1<br>Bit 1: Port size = 2<br>Bit 2: Port size = 3, not supported<br>Bit 3: Port Size = 4<br>Only 1 bit must be set. By default, the port size is one bit. (0x00000001) |
| 0xE00400F0 | Selected pin protocol | Allows the Trace Port Protocol to be selected:<br>Bit1:0=<br>00: Sync Trace Port Mode<br>01: Serial Wire Output - manchester (default value)<br>10: Serial Wire Output - NRZ<br>11: reserved                                            |

Table 150. Important TPIU registers (continued)

| Address    | Register                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0xE0040304 | Formatter and flush control | <p>Bits 31-9 = always '0</p> <p>Bit 8 = TrgIn = always '1 to indicate that triggers are indicated</p> <p>Bits 7-4 = always 0</p> <p>Bits 3-2 = always 0</p> <p>Bit 1 = EnFCont. In Sync Trace mode (Select_Pin_Protocol register bit1:0=00), this bit is forced to '1: the formatter is automatically enabled in continuous mode. In asynchronous mode (Select_Pin_Protocol register bit1:0 &lt;&gt; 00), this bit can be written to activate or not the formatter.</p> <p>Bit 0 = always 0</p> <p>The resulting default value is 0x102</p> <p><b>Note:</b> In synchronous mode, because the TRACECTL pin is not mapped outside the chip, the formatter is always enabled in continuous mode -this way the formatter inserts some control packets to identify the source of the trace packets).</p> |
| 0xE0040300 | Formatter and flush status  | Not used in Cortex®-M4 with FPU, always read as 0x00000008                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

### 23.17.10 Example of configuration

- Set the bit TRCENA in the Debug Exception and Monitor Control Register (DEMCR)
- Write the TPIU Current Port Size Register to the desired value (default is 0x1 for a 1-bit port size)
- Write TPIU Formatter and Flush Control Register to 0x102 (default value)
- Write the TPIU Select Pin Protocol to select the sync or async mode. Example: 0x2 for async NRZ mode (UART like)
- Write the DBGMCU control register to 0x20 (bit IO\_TRACEN) to assign TRACE I/Os for async mode. A TPIU Sync packet is emitted at this time (FF\_FF\_FF\_7F)
- Configure the ITM and write the ITM Stimulus register to output a value

## 23.18 DBG register map

The following table summarizes the Debug registers.

**Table 151. DBG register map and reset values**

| Addr.          | Register                   | 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17            | 16 | 15       | 14 | 13       | 12 | 11     | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1               | 0 |   |   |            |   |   |   |          |  |  |  |             |   |   |   |          |   |   |   |           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   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|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   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|----------------|----------------------------|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|---------------|----|----------|----|----------|----|--------|----|---|---|---|---|---|---|---|---|-----------------|---|---|---|------------|---|---|---|----------|--|--|--|-------------|---|---|---|----------|---|---|---|-----------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--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| 0xE004<br>2000 | DBGMCU_IDCODE              | REV_ID   |    |    |    |    |    |    |    |    |    |    |    |    |    |               |    | Reserved |    |          |    | DEV_ID |    |   |   |   |   |   |   |   |   |                 |   |   |   |            |   |   |   |          |  |  |  |             |   |   |   |          |   |   |   |           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   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|                | Reset value <sup>(1)</sup> | X        | X  | X  | X  | X  | X  | X  | X  | X  | X  | X  | X  | X  | X  | X             | X  |          |    |          |    | X      | X  | X | X | X | X | X | X | X | X | X               | X | X | X | X          |   |   |   |          |  |  |  |             |   |   |   |          |   |   |   |           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   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|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   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| 0xE004<br>2004 | DBGMCU_CR                  | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    | DBG_TIM5_STOP |    |          |    | Reserved |    |        |    |   |   |   |   |   |   |   |   | TRACE_MODE[1:0] |   |   |   | TRACE_IOEN |   |   |   | Reserved |  |  |  | DBG_STANDBY |   |   |   | DBG_STOP |   |   |   | DBG_SLEEP |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   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|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|                | Reset value                |          |    |    |    |    |    |    |    |    |    |    |    |    |    | 0             | 0  | 0        | 0  |          |    |        |    |   |   |   |   |   |   |   |   | 0               | 0 | 0 | 0 | 0          | 0 | 0 | 0 |          |  |  |  | 0           | 0 | 0 | 0 | 0        | 0 | 0 | 0 | 0         | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

1. The reset value is product dependent. For more information, refer to [Section 23.6.1: MCU device ID code](#).

## 24 Device electronic signature

The electronic signature is stored in the Flash memory area. It can be read using the JTAG/SWD or the CPU. It contains factory-programmed identification data that allow the user firmware or other external devices to automatically match its interface to the characteristics of the STM32F4xx microcontrollers.

### 24.1 Unique device ID register (96 bits)

The unique device identifier is ideally suited:

- for use as serial numbers (for example USB string serial numbers or other end applications)
- for use as security keys in order to increase the security of code in Flash memory while using and combining this unique ID with software cryptographic primitives and protocols before programming the internal Flash memory
- to activate secure boot processes, etc.

The 96-bit unique device identifier provides a reference number which is unique for any device and in any context. These bits can never be altered by the user.

The 96-bit unique device identifier can also be read in single bytes/half-words/words in different ways and then be concatenated using a custom algorithm.

**Base address: 0x1FFF 7A10**

Address offset: 0x00

Read only = 0xXXXX XXXX where X is factory-programmed

|           |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|-----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31        | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| UID(31:0) |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| r         | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r | r | r | r | r | r | r | r | r | r |

Bits 31:0 **UID(31:0)**: X and Y coordinates on the wafer.

Address offset: 0x04

Read only = 0xXXXX XXXX where X is factory-programmed

|            |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 31         | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| UID(63:48) |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| r          | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  |
| 15         | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| UID(47:32) |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| r          | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  |

Bits 31:8 **UID(63:40):** LOT\_NUM[23:0]  
Lot number (ASCII encoded).

Bits 7:0 **UID(39:32):** WAF\_NUM[7:0]  
Wafer number (ASCII encoded).

Address offset: 0x08

Read only = 0xFFFF XXXX where X is factory-programmed

|            |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 31         | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| UID(95:80) |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| r          | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  |
| 15         | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| UID(79:64) |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| r          | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  |

Bits 31:0 **UID(95:64):** LOT\_NUM[55:24]  
Lot number (ASCII encoded).

## 24.2 Flash size

Base address: 0x1FFF 7A22

Address offset: 0x00

Read only = 0xFFFF where X is factory-programmed

|        |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|--------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 15     | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| F_SIZE |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| r      | r  | r  | r  | r  | r  | r | r | r | r | r | r | r | r | r | r |

Bits 15:0 **F\_ID(15:0):** Flash memory size  
This bitfield indicates the size of the device Flash memory expressed in Kbytes.  
As an example, 0x0400 corresponds to 1024 Kbytes.

## 25 Revision history

**Table 152. Document revision history**

| Date         | Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 24-July-2014 | 1       | Initial version                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 15-Sep-2017  | 2       | <p>Added:</p> <ul style="list-style-type: none"> <li>– <a href="#">Section 5.3: Low-power modes</a></li> <li>– <a href="#">Exiting low-power mode</a></li> <li>– <a href="#">Table 63: Minimum and maximum timeout values at 30 MHz (<math>f_{PCLK1}</math>)</a></li> <li>– <a href="#">Table 132: TRDT values</a></li> </ul> <p>Updated:</p> <ul style="list-style-type: none"> <li>– <a href="#">Table 8: Option byte organization</a></li> <li>– <a href="#">Table 14: Low-power mode summary</a></li> <li>– <a href="#">Section 4.2: CRC main features</a></li> <li>– <a href="#">Section 5.4.1: PWR power control register (PWR_CR)</a></li> <li>– <a href="#">Entering Sleep mode, Exiting Sleep mode</a></li> <li>– <a href="#">Table 15: Sleep-now entry and exit</a></li> <li>– <a href="#">Table 16: Sleep-on-exit entry and exit</a></li> <li>– <a href="#">Entering Stop mode</a></li> <li>– <a href="#">Exiting Stop mode</a></li> <li>– <a href="#">Entering Standby mode, Exiting Standby mode</a></li> <li>– <a href="#">Section 8.3.6: GPIO locking mechanism</a></li> <li>– <a href="#">Section 8.4.1: GPIO port mode register (GPIOx_MODER) (x = A..E and H)</a></li> <li>– <a href="#">Section 11.12.7: ADC watchdog higher threshold register (ADC_HTR)</a></li> <li>– <a href="#">Section 11.12.8: ADC watchdog lower threshold register (ADC_LTR)</a></li> <li>– <a href="#">Section 12.4.12: TIM1 auto-reload register (TIMx_ARR)</a></li> <li>– <a href="#">Section 13.4.10: TIMx counter (TIMx_CNT)</a></li> <li>– <a href="#">Section 13.4.10: TIMx counter (TIMx_CNT)</a></li> <li>– <a href="#">Section 13.4.11: TIMx prescaler (TIMx_PSC)</a></li> <li>– <a href="#">Section 13.4.12: TIMx auto-reload register (TIMx_ARR)</a></li> <li>– <a href="#">Section 13.4.15: TIMx capture/compare register 3 (TIMx_CCR3)</a></li> <li>– <a href="#">Section 14.4.5: TIM9 event generation register (TIMx_EGR)</a></li> <li>– <a href="#">Section 16.4: How to program the watchdog timeout</a></li> <li>– <a href="#">Section 23.15.4: ETM configuration example</a></li> <li>– <a href="#">Section 23.16.2: Debug support for timers, watchdog and I<sup>2</sup>C</a></li> <li>– <a href="#">Table 18: Stop mode entry and exit</a></li> <li>– <a href="#">Table 19: Standby mode entry and exit</a></li> <li>– <a href="#">Table 53: TIMx internal trigger connection</a></li> <li>– <a href="#">Table 53: TIMx internal trigger connection</a></li> <li>– <a href="#">Figure 157: Watchdog block diagram</a></li> </ul> |
| 30-Nov-2018  | 3       | <p>Updated:</p> <ul style="list-style-type: none"> <li>– <a href="#">Section 9.3.4: Arbiter</a></li> <li>– <a href="#">Section 22: USB on-the-go full-speed (OTG_FS)</a></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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